

SYSTEM 4.0
04-S28-2014
02/09/15

APPROVALS		 MOTOROLA MOBILE DEVICES	
ENGINEER: MOTOROLA MOBILITY		BLOCK TITLE:	
LAST MODIFIED BY: NATE CONNELL Fri Dec 18 13:27:05 2015		84016530001_a	
MOTOROLA CONFIDENTIAL RESTRICTED		PROJECT NAME	
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		PAGE 1 OF 52	

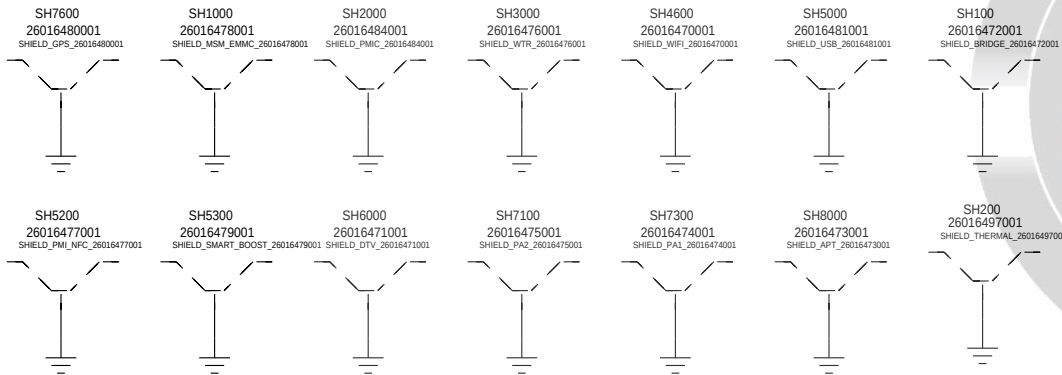
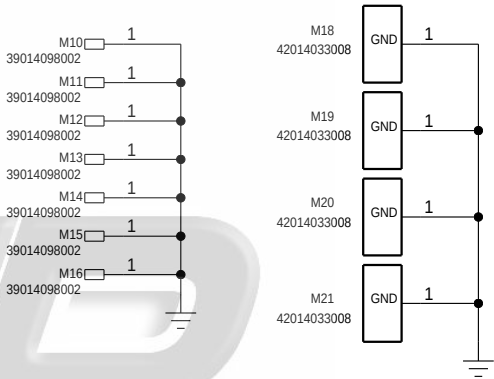


AFFINITY ME/SHIELDS

AFFINITY REFERENCE DESIGNATOR RANGES BY BLOCK

- 000 - 499: ANTENNA
- 500 - 599: USER INPUT
- 600 - 799: POWER
- 1000-2999: MSM8852 + PM8952
- 3000-3049, 3100-3149: WTR2955
- 3050-3099, 3150-3199: CA_WTR2955
- 3200-3999: MSM8952 + PM8952
- 4000-4499: CONNECTIVITY
- 4500-4999: AUDIO FE
- 5000-5499: DISPLAY
- 5500-5999: CAMERA
- 6000-6999: XMCS
- 7000-8999: RFFE
- 9000-9999: PRODUCT-SPECIFIC (FILTERING,, 100PF CAPS)

UNIVERSAL GROUND CONTACTS



MOTOROLA MOBILITY CONFIDENTIAL RESTRICTED-MDB TRACKED TRACKEDANDBLOCKED

APPROVALS		MOTOROLA MOBILE DEVICES							
ENGINEER:		BLOCK TITLE:							
LAST MODIFIED BY: Fri Feb 19 14:26:03 2016		84016530001_a							
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B	PROJECT NAME 84016530001_a.cpm	REV 01							
CAD FILE NAME:		PAGE 2 OF 52							

[illegible]

The schematic diagram illustrates the RF front end for GSM/UTMS/LTE, specifically for the P0_112 room. It shows the signal path from the antenna through various filters and amplifiers to the baseband processor.

Key Components and Connections:

- Antenna:** DIV_ANT_QFE_OUT (1100PF, 2113944E23) connected to the antenna input.
- Filters:**
 - DRX_LB_ASM_RFOUT (IN) and DRX_LB_ASM_RFOUT (IN) are connected to the antenna input.
 - DRX_LB_DIP (1100PF, 2113944E17) is a diplexer filter.
 - DRX_LB_HB_DIP (1100PF, 2113944E17) is a diplexer filter.
 - DRX_LB_HB_RFCON_IN (1100PF, 2113944E23) is a filter.
 - DRX_LB_HB_RFCON_OUT (1100PF, 2113944E23) is a filter.
 - DRX_LB_HB_FTPOUT (1100PF, 2113944E23) is a filter.
- Amplifiers:**
 - 91859006 (LFD21892MDPCC065) is a low noise amplifier (LNA).
 - 0613952Y66 (ESD) is an ESD protection diode.
- Other Components:**
 - DRX_LB_HB_DIP (1100PF, 2113944E17) is a diplexer filter.
 - DRX_LB_HB_RFCON_IN (1100PF, 2113944E23) is a filter.
 - DRX_LB_HB_RFCON_OUT (1100PF, 2113944E23) is a filter.
 - DRX_LB_HB_FTPOUT (1100PF, 2113944E23) is a filter.

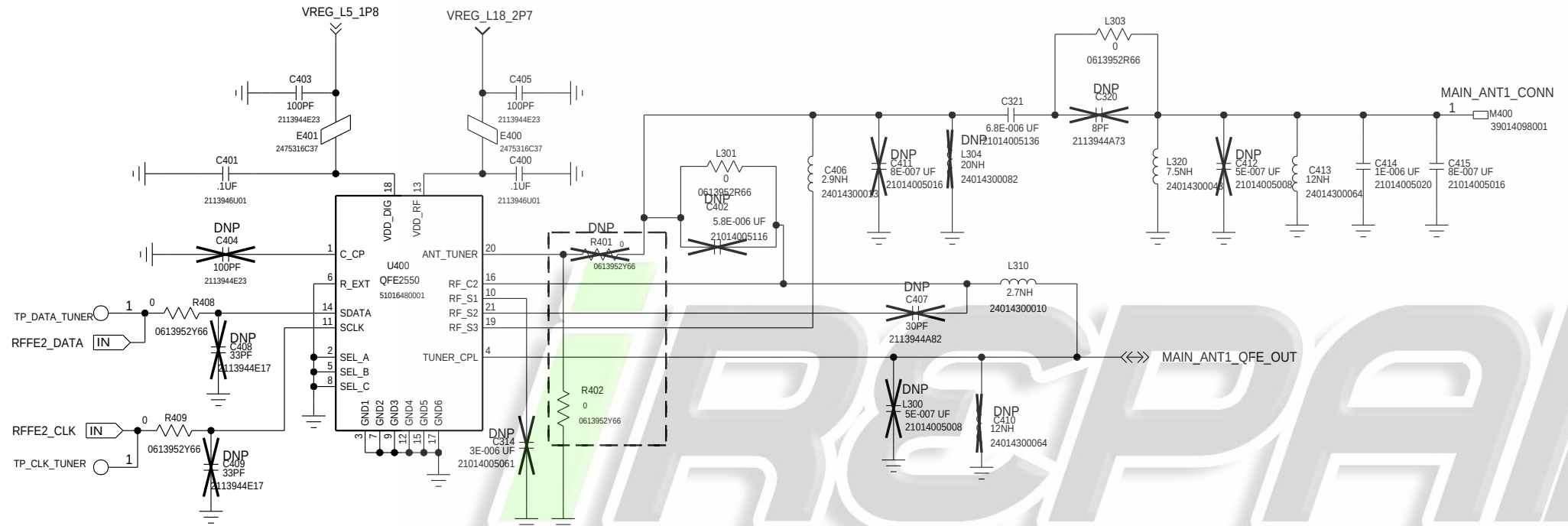
The diagram shows the signal path from the antenna through various filters and amplifiers to the baseband processor. The components are labeled with their part numbers and values, and the connections are shown with lines and arrows.

[illegible]PAGE 3 of 52

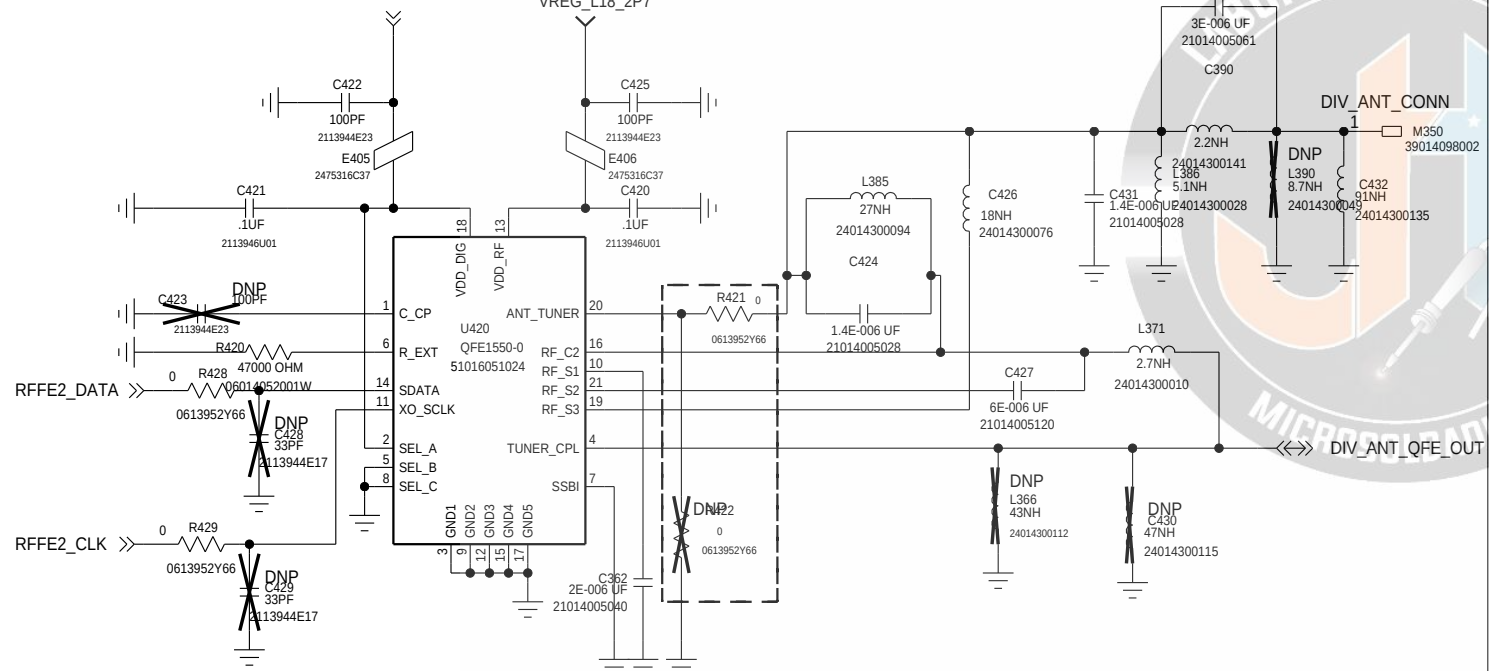
ANT: QCOM TUNERS

REFDES: 400-499

MAIN 1



DIV

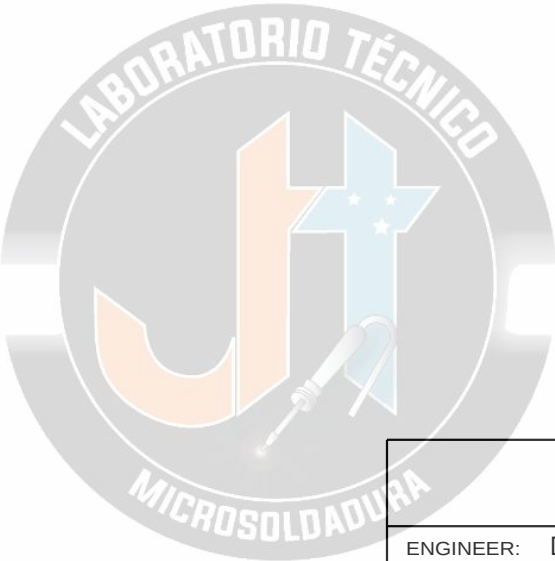
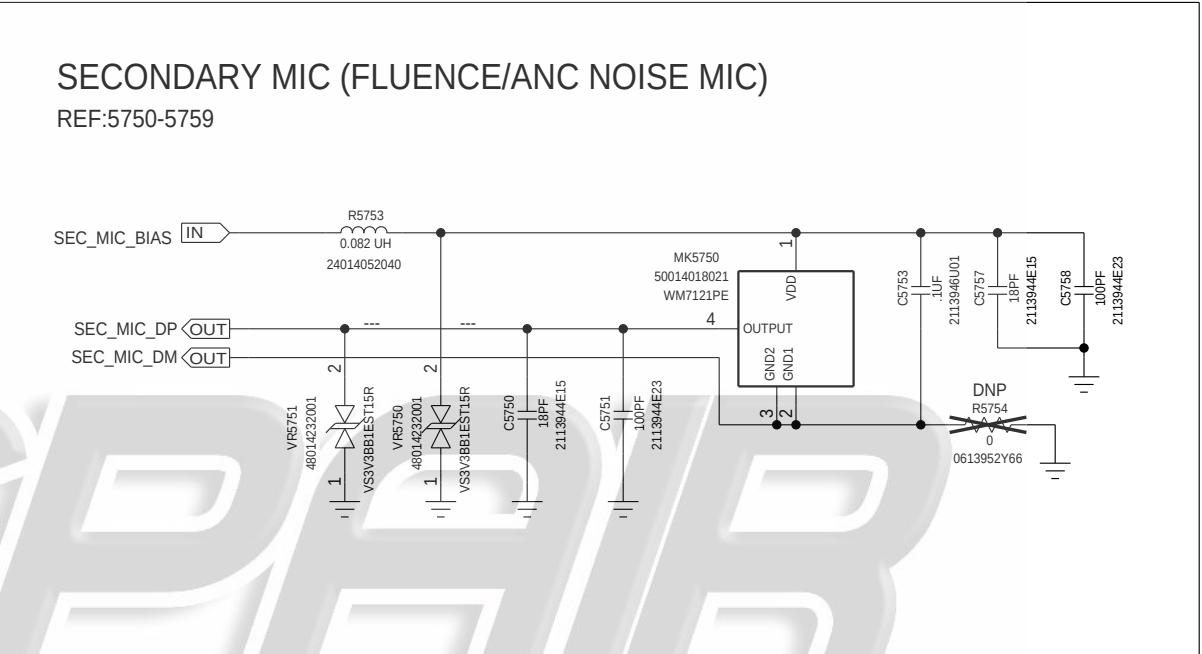
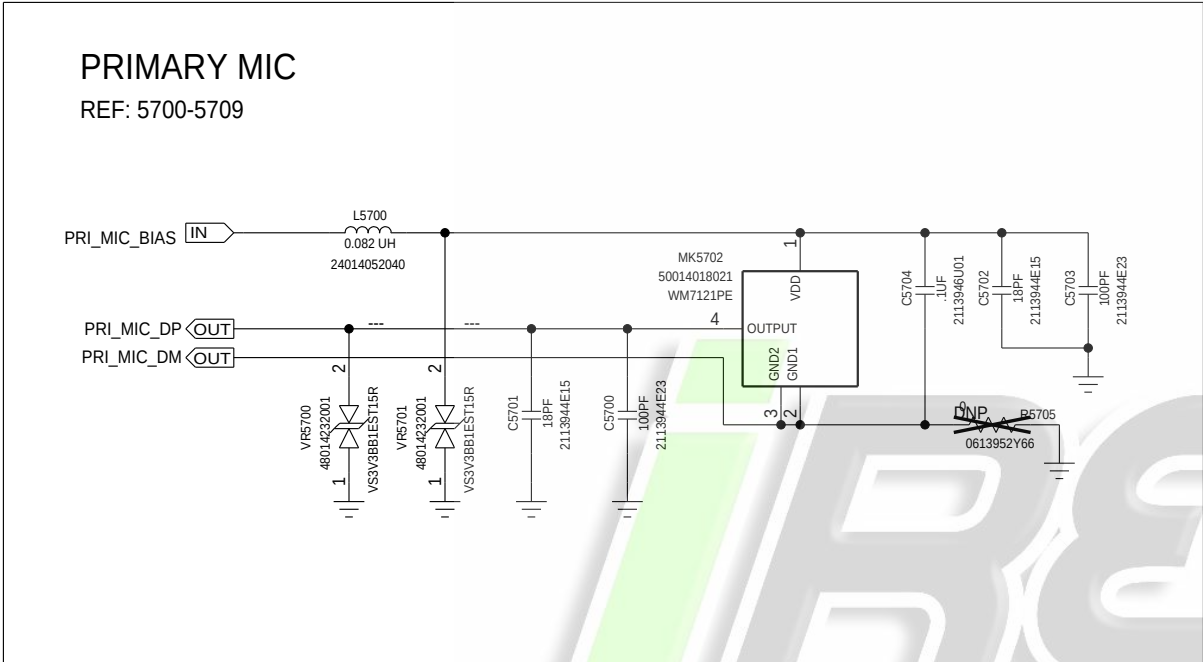


U420 REPLACE TO QFE1550

APPROVALS		 MOTOROLA MOBILE DEVICES	
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		PAGE 4 OF 52	

AFE: MICS

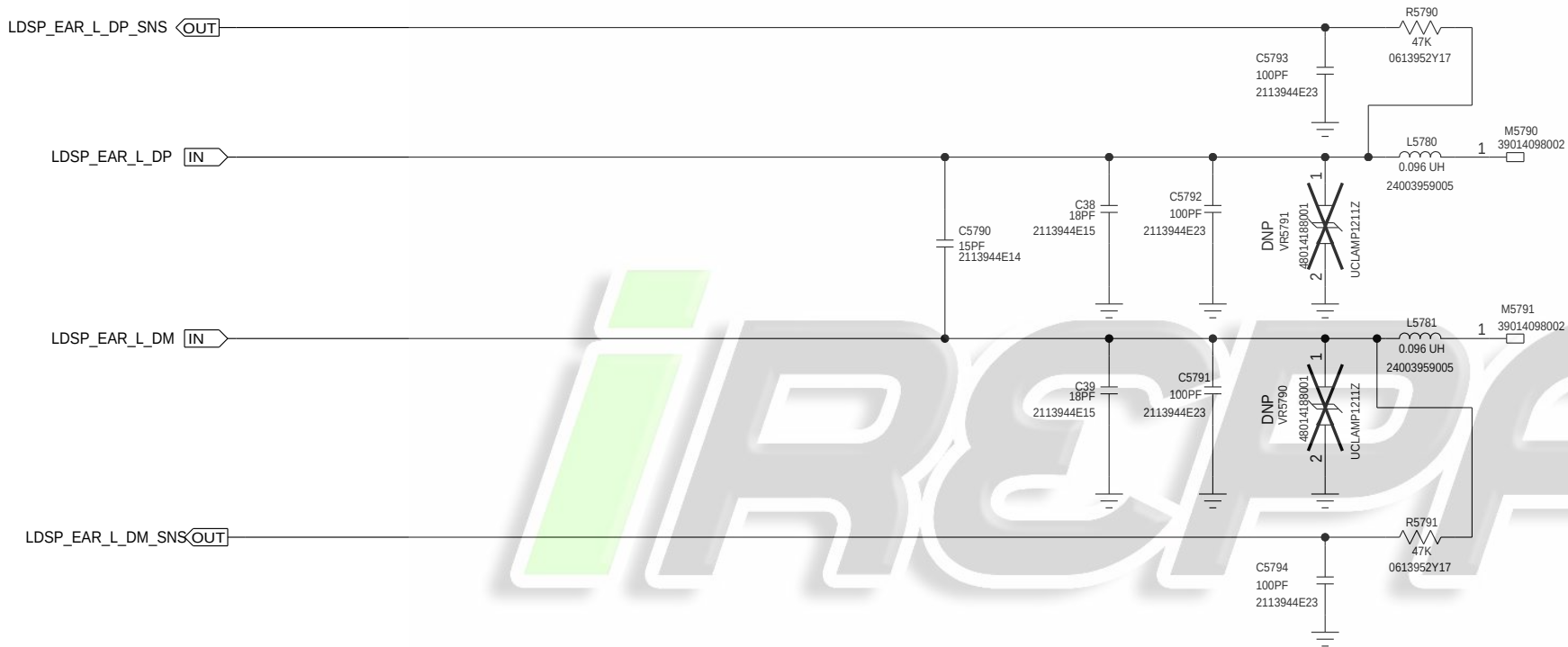
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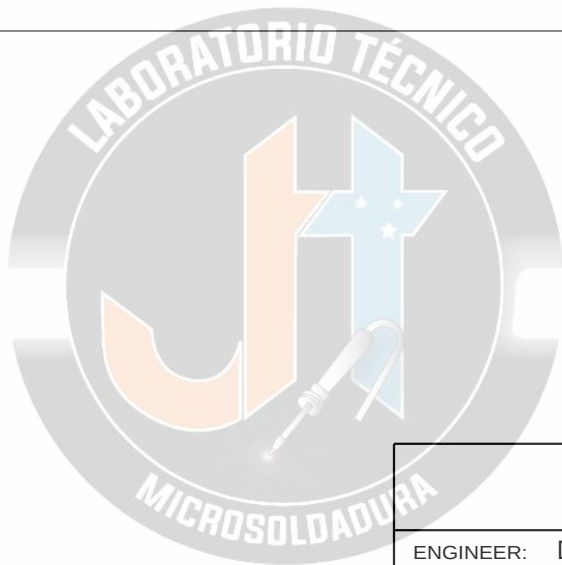
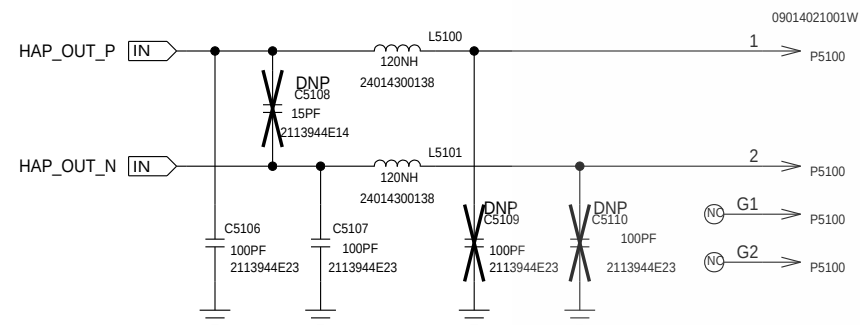
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		CAD FILE NAME:	
		PAGE 5 OF 52	

AFE: SPEAKERS

LOUDSPEARKER/EAR PIECE
REF:5790-5799



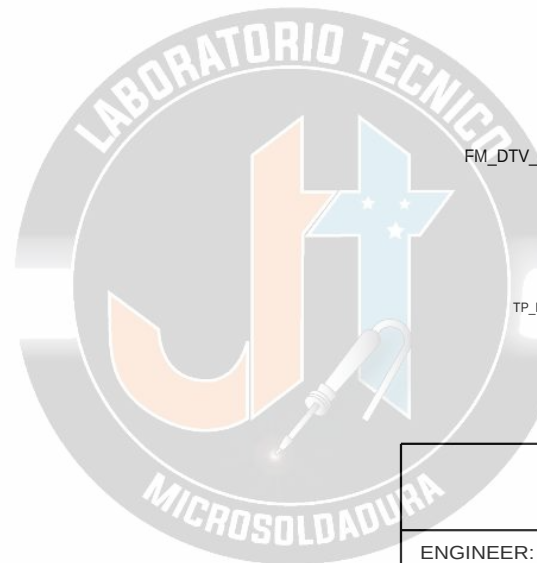
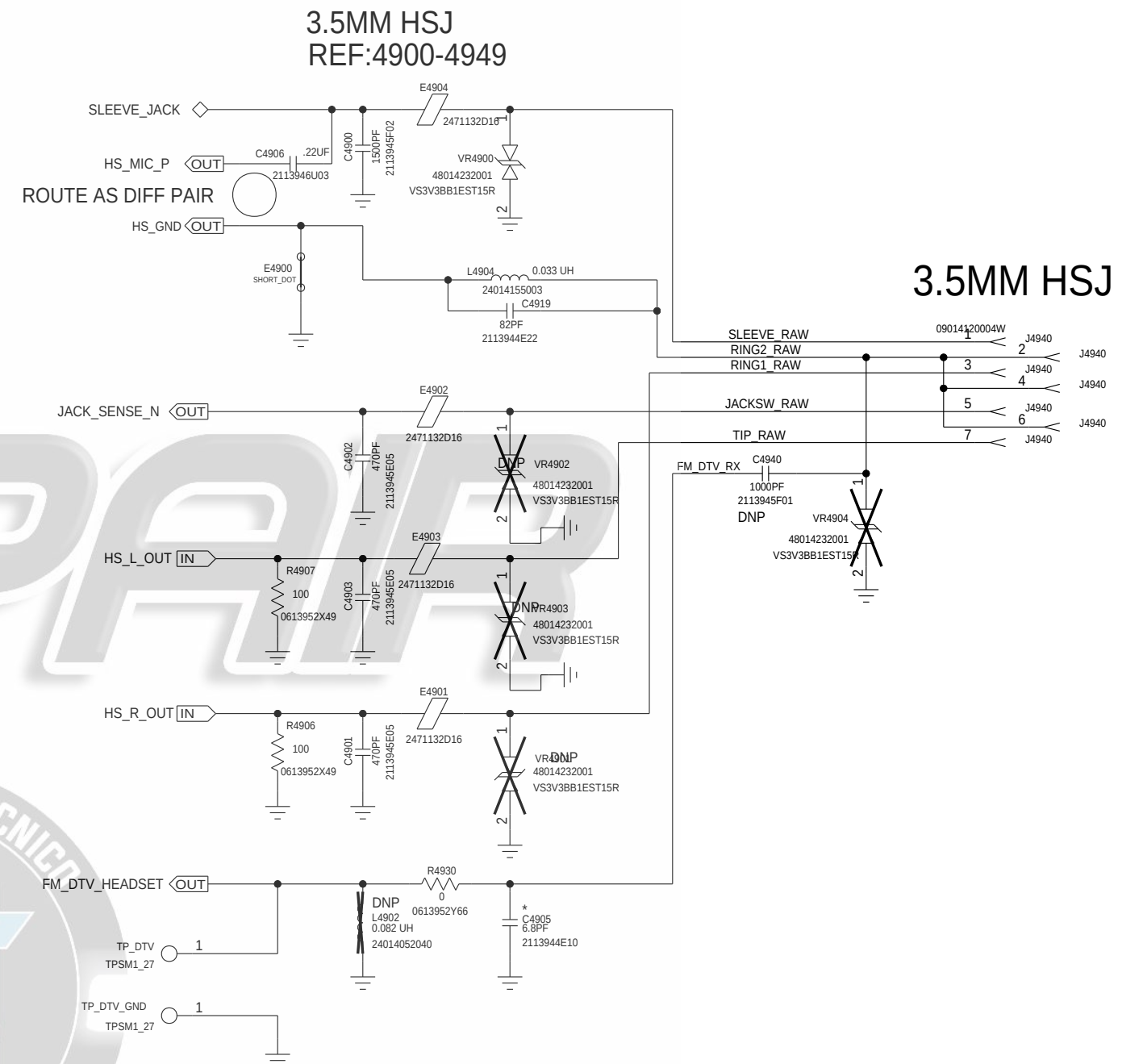
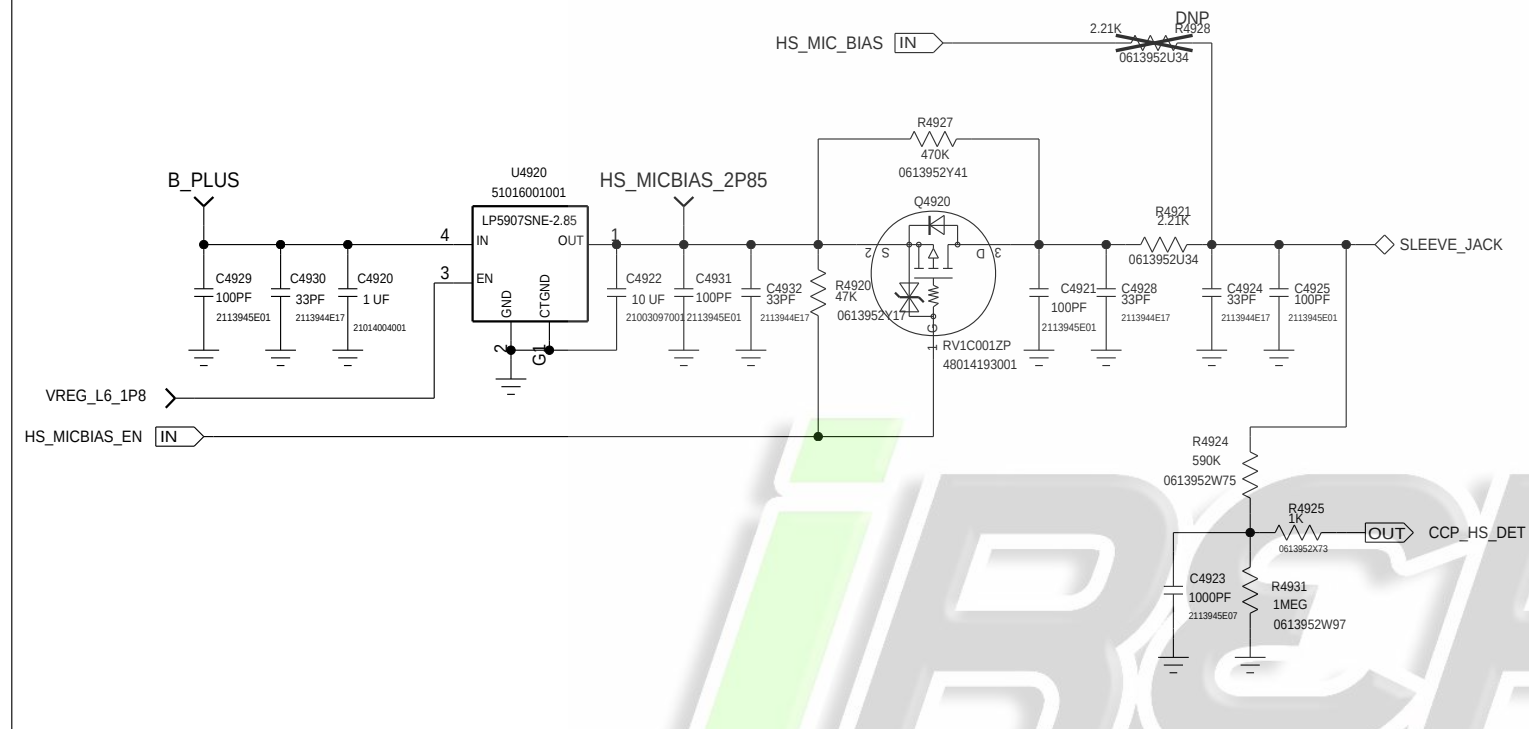
ROTARY VIBRATOR
REF 5100-5119



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			REV 01 PAGE 6 OF 52

AFE: 3.5MM HEADSET

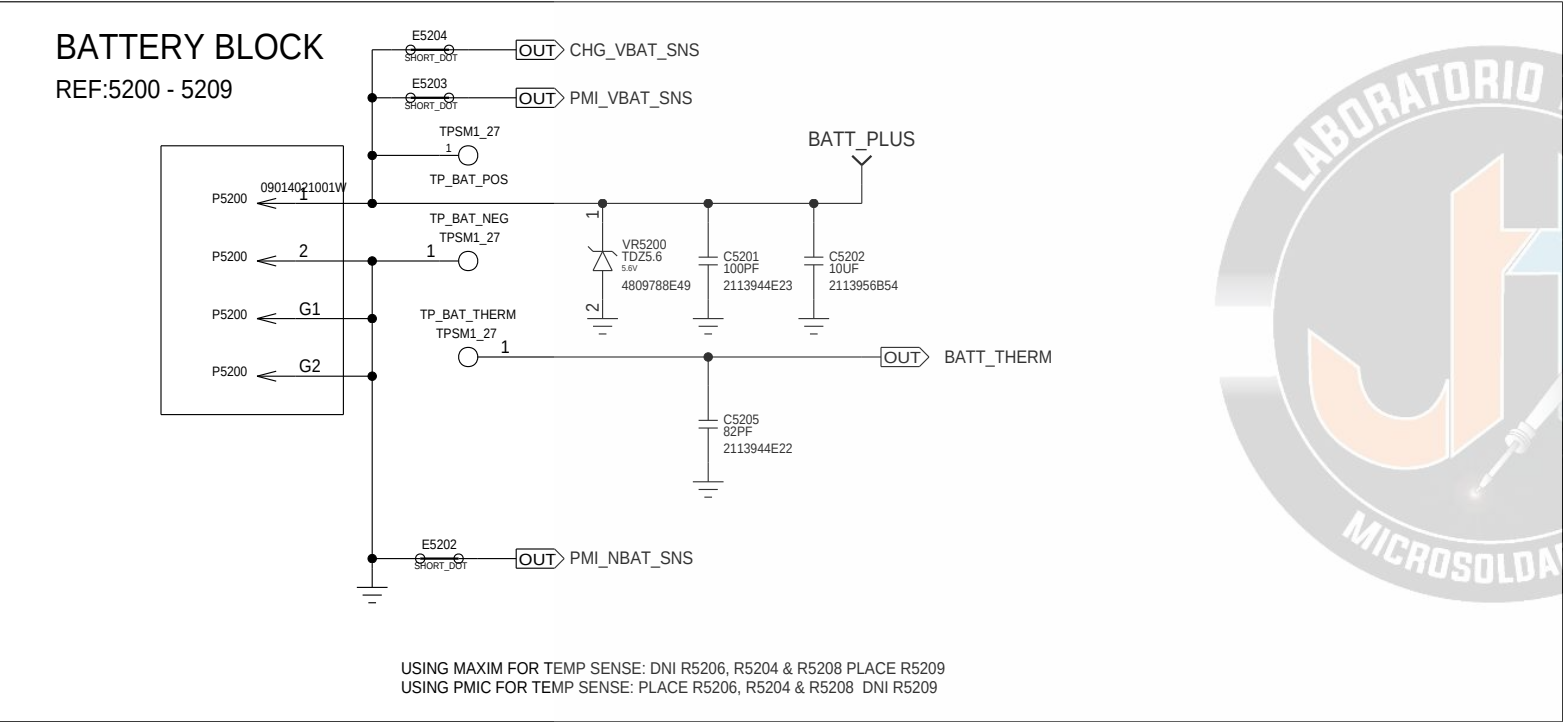
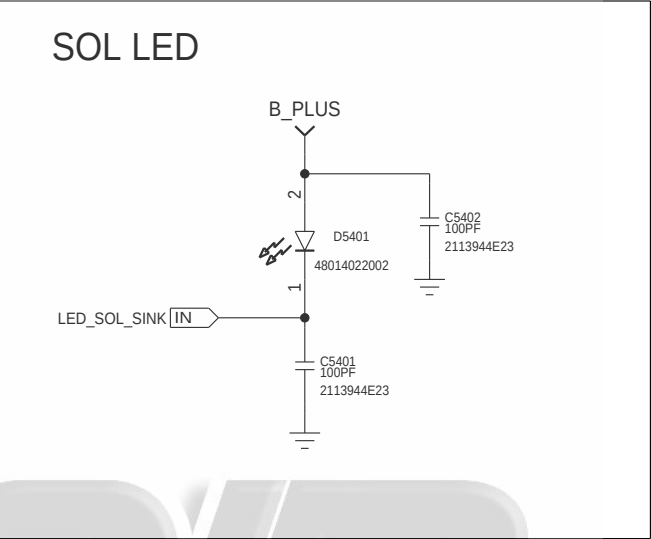
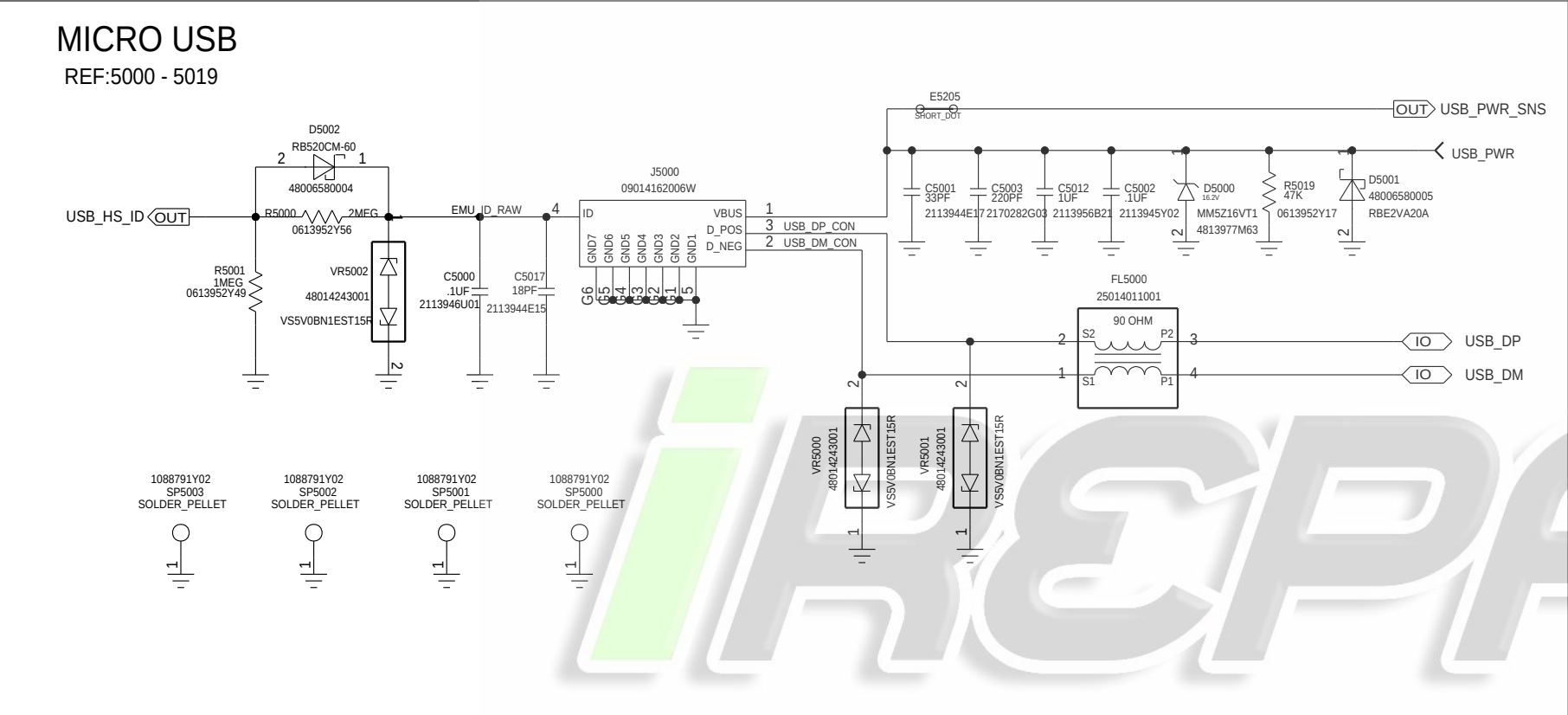
SENSOR HUB HEADSET DETECTION CIRCUIT



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PWR: USB2 / BATT

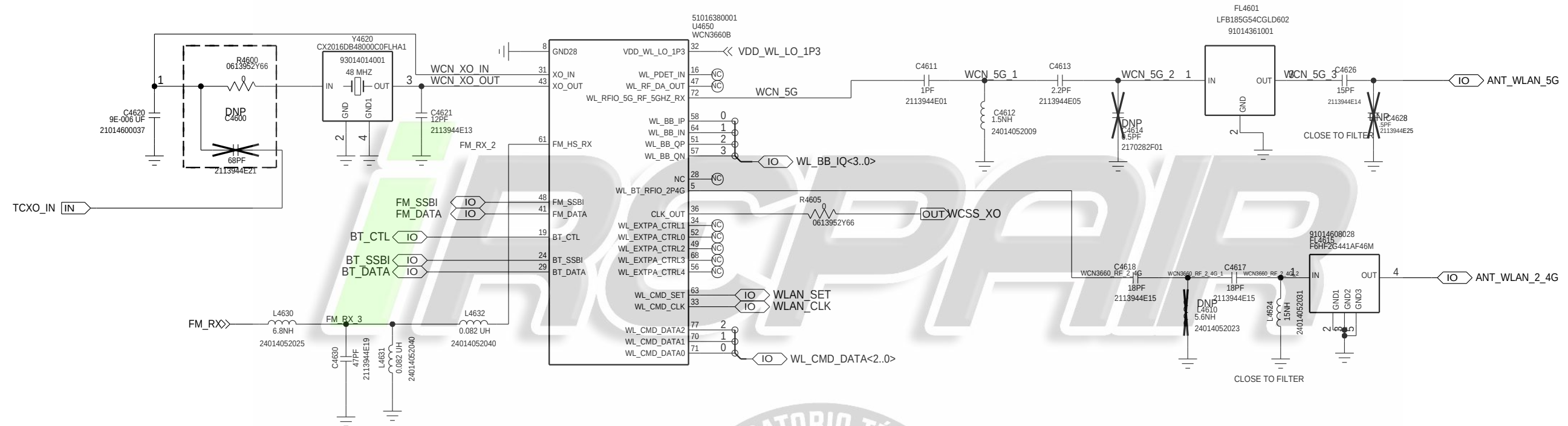
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WCN3660B RF/CTRL
REF 4600-4699

NEED TO CHECK PIN47 CONNECTION



2.4G ONLY: PLACE-C4600, DNP-Y4620, C4620, C4621, R4600, R4605, ALL 5G RF PATH
2.4G/5G: DNP-C4600, PLACE-Y4620, C4621, R4600, R4605, ALL 5G RF PATH

BASED ON 80-WL006-44 REV C

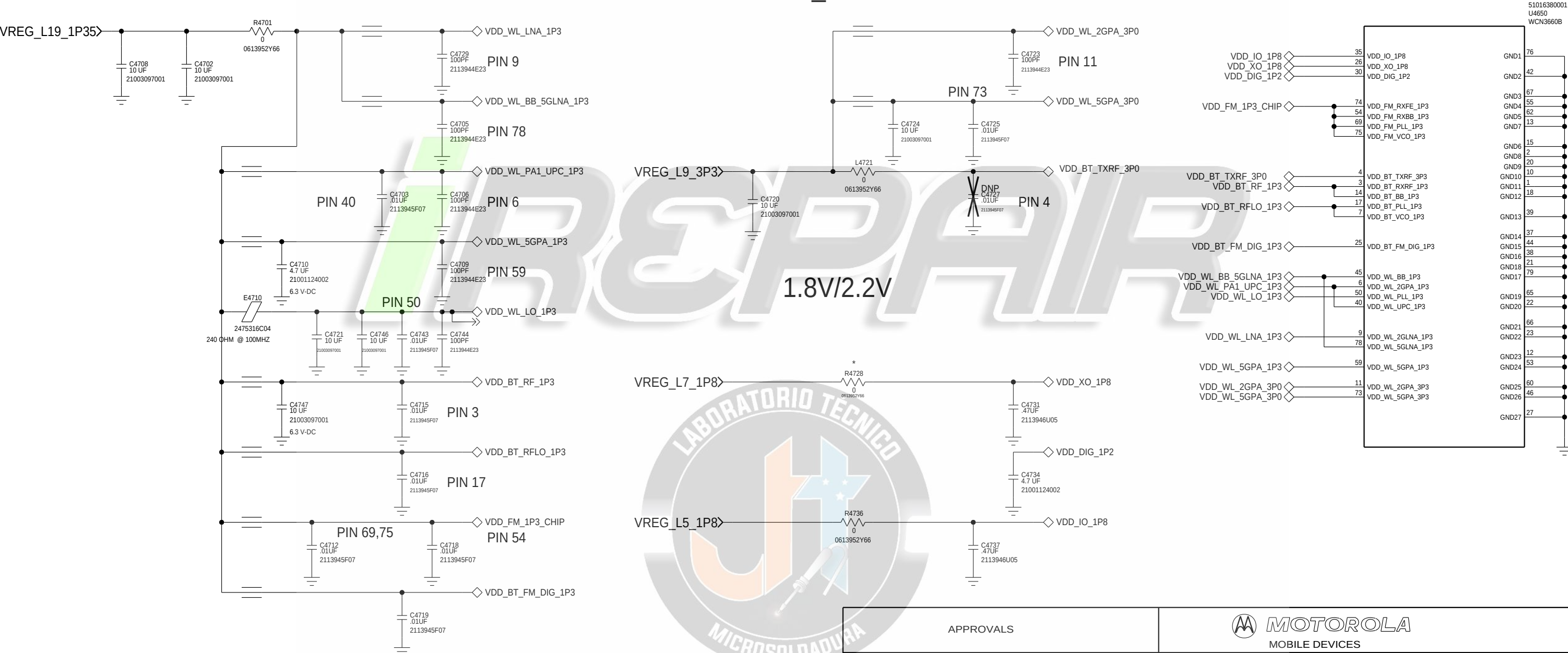
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LAST MODIFIED BY: FROEHLING Fri Feb 19 14:26:12 2016			
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WCN3660B POWER
REF 4700-4799

NEED TO SCRUB ALL SUPPLY SEQUENCE/VOLTAGE/CURRENT NUMBERS FOR WCN3660B

BT_FM_WLAN_1P3

WLBT_3.0V

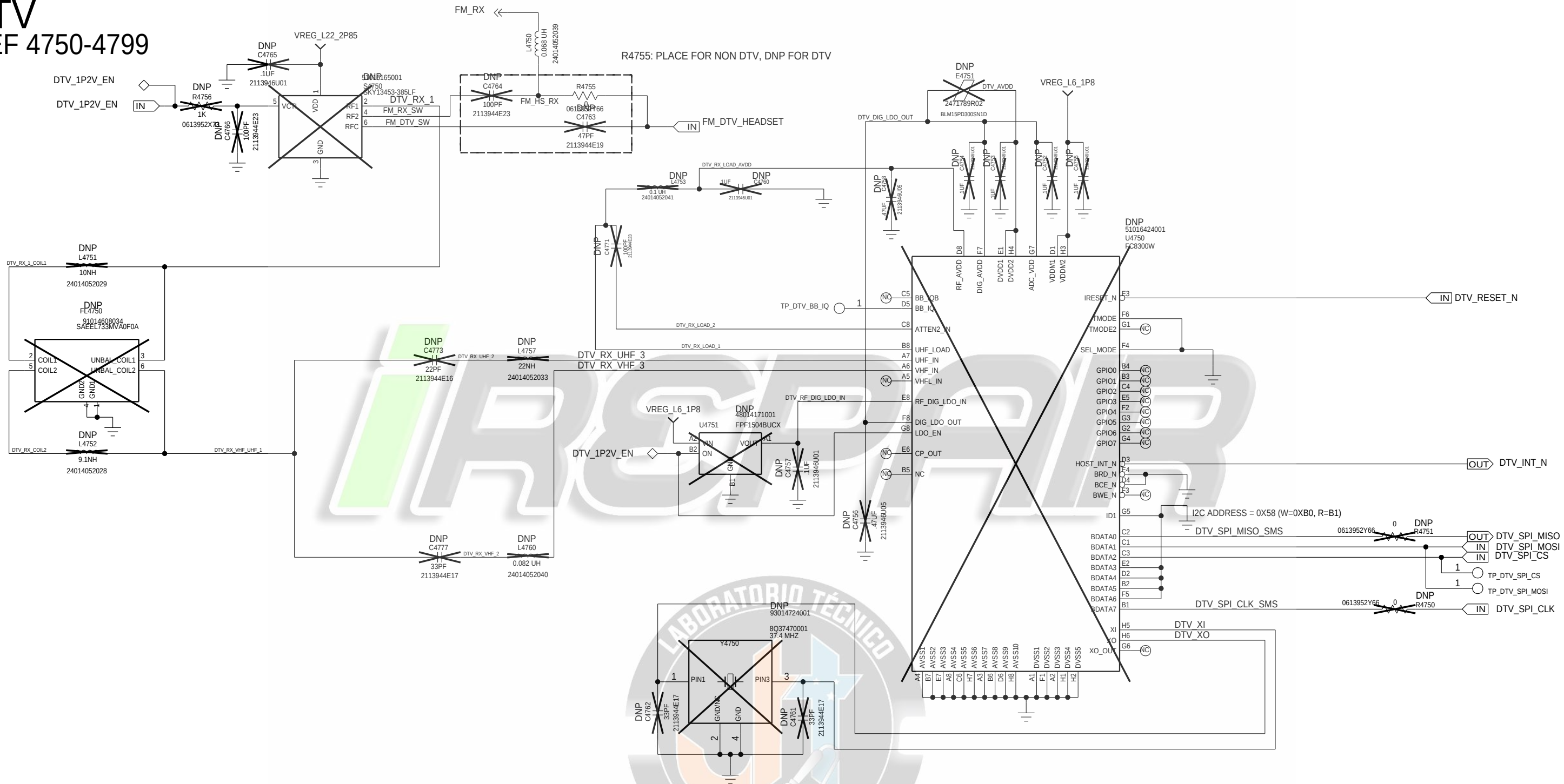


NOTE: RESERVE C4717 AND C4718 FOR 0.01UF BYPASSES

BASED ON 80-WL006-44 REV C

APPROVALS		MOTOROLA MOBILE DEVICES	
ENGINEER:		BLOCK TITLE:	
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		CAD FILE NAME:	PAGE 10 OF 52

DTV
REF 4750-4799



APPROVALS		MOTOROLA MOBILE DEVICES	
ENGINEER:		BLOCK TITLE:	
LAST MODIFIED BY: GSHIMON Fri Feb 19 14:26:11 2016		connectivity	
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		B 84016530001_a.cpm	01
CAD FILE NAME:		PAGE 11 OF 52	

CONN: UIM 1& 2

UIM1 CONN
REF:5500-5509

UIM_PRESENT:
HIGH WHEN CARD PRESENT (OPEN)
LOW WHEN CARD ABSENT (CLOSED)

VREG_L5_1P8 VREG_L14_UIM1

UIM1_POWER VCC_SIM1

UIM1_SWP <>>
UIM1_DATA IO
UIM1_RESET IN
UIM1_CLK IN
UIM1_PRESENT OUT

R5502 47K 0613952Y17 R5501 15K 0613952Y05

R5504 100 0613952X49 R5503 100 0613952X49 R5506 1K 0613952X73

DNP C5500 100PF 2113944E23 DNP C5505 33PF 2113944E17 DNP C5502 33PF 2113944E17 DNP C5503 33PF 2113944E17 VR5500 VS3V3B81EST15R 48014232001 2 VS5501 VS3V3B81EST15R 48014232001 2 VR5502 VS3V3B81EST15R 48014232001 2 VR5503 VS3V3B81EST15R 48014232001 2 VR5505 VS3V3B81EST15R 48014232001 2

J5500 39014196001 CONN_J DETECT_B DB DETECT_A G6

C1 C6 C7 I_O RESERVED1 RESERVED2 C4 CB DB G6

GND G1 G2 G3 G4 G5 G6 G7

UIM2 CONN
REF:5520-5529

UIM_PRESENT:
HIGH WHEN CARD PRESENT (OPEN)
LOW WHEN CARD ABSENT (CLOSED)

VREG_L5_1P8 VREG_L15_UIM2

UIM2_POWER VCC_SIM2

UIM2_SWP <>>
UIM2_DATA IO
UIM2_RESET IN
UIM2_CLK IN
UIM2_PRESENT OUT

R5522 47K 0613952Y17 R5521 15K 0613952Y05

R5524 100 0613952X49 R5523 100 0613952X49 R5526 1K 0613952X73

DNP C5524 100PF 2113944E23 DNP C5522 33PF 2113944E17 DNP C5523 33PF 2113944E17 DNP C5525 33PF 2113944E17 VR5520 VS3V3B81EST15R 48014232001 2 VS5521 VS3V3B81EST15R 48014232001 2 VR5522 VS3V3B81EST15R 48014232001 2 VR5523 VS3V3B81EST15R 48014232001 2 VR5525 VS3V3B81EST15R 48014232001 2

J5520 39014196001 CONN_J DETECT_B DB DETECT_A G6

C1 C6 C7 I_O RESERVED1 RESERVED2 C4 CB DB G6

GND G1 G2 G3 G4 G5 G6 G7

APPROVALS

ENGINEER: FROEHLING

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MOTOROLA
MOBILE DEVICES

BLOCK TITLE:
connectivity

PROJECT NAME
84016530001_a.cpm

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CAD FILE NAME:

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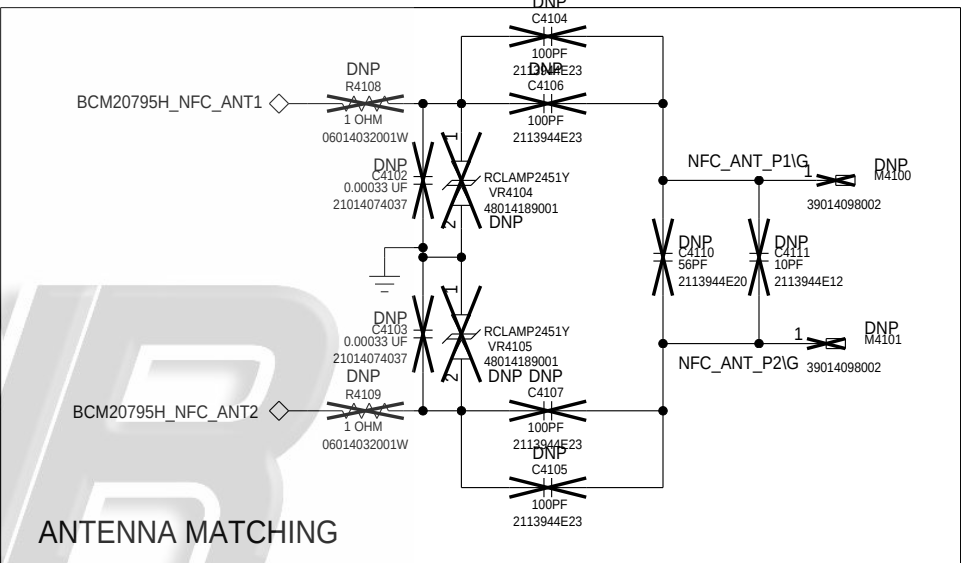
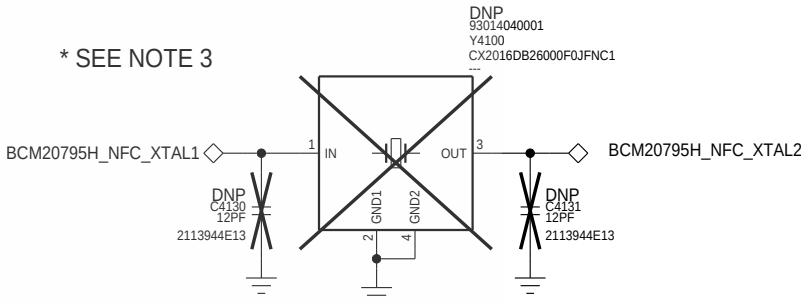
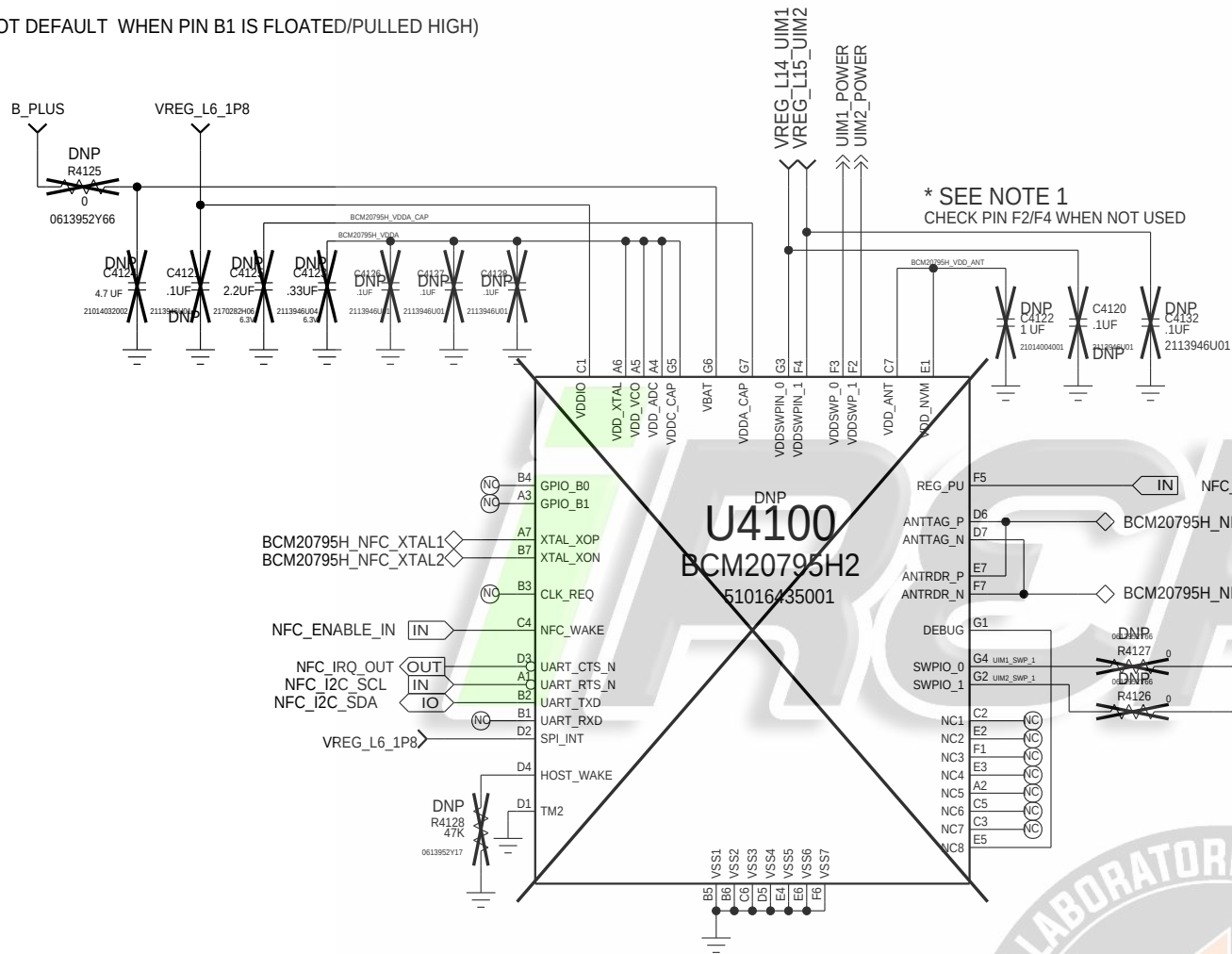
BROADCOM NFC

REF: 4100-4199

I2C ADDR: 0X77 (BOOT DEFAULT WHEN PIN B1 IS PULLED TO GROUND)

OR

I2C ADDR: 0X76 (BOOT DEFAULT WHEN PIN B1 IS FLOATED/PULLED HIGH)

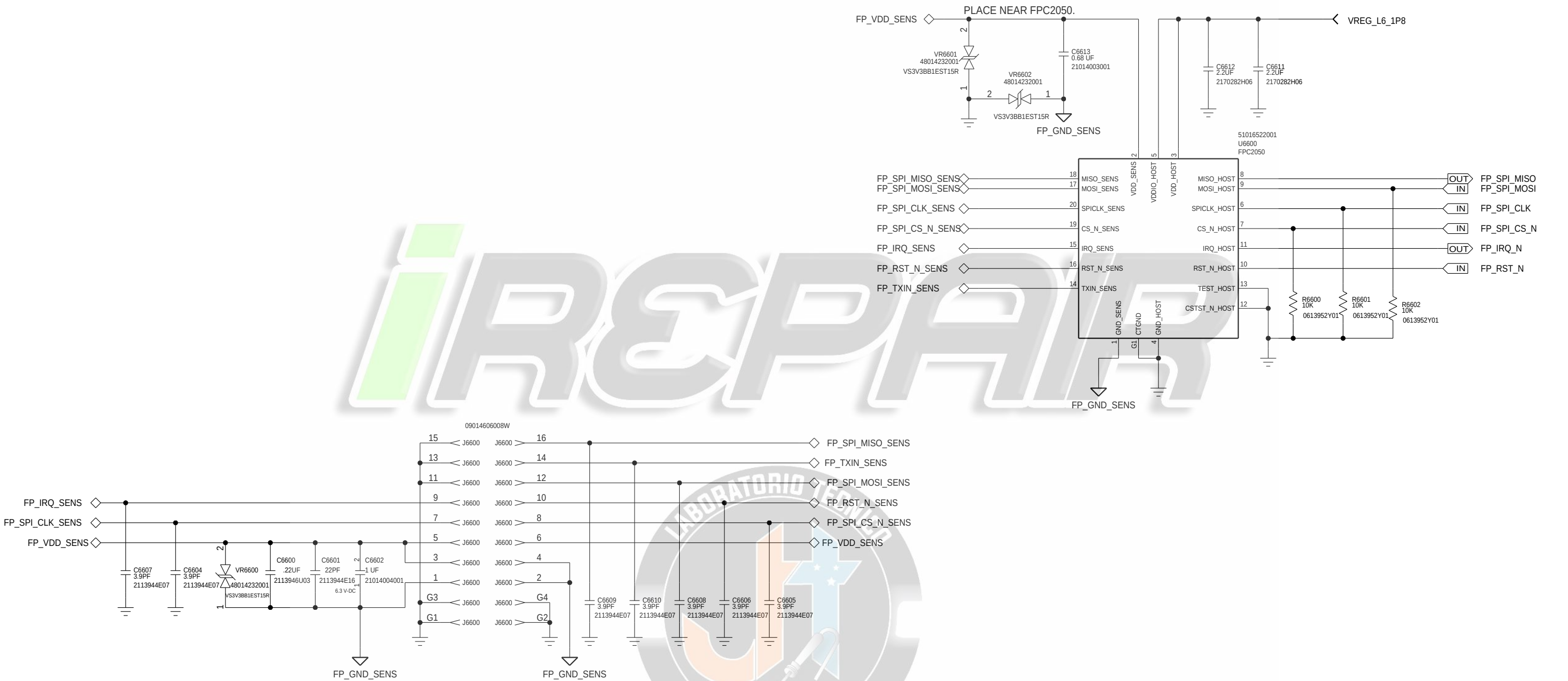


NOTES:

1. PER BROADCOM RECOMMENDATION, CAPACITANCE ON UIM1_POWER NET SHOULD BE 0.33UF. TOO LITTLE COMPROMISES REGULATOR STABILITY, TOO MUCH CAN CAUSE INADVERTENT POWER-ON RESETS AT VDD.
2. PLACE U4101 AND C4100 ONLY ON CHINA SKU
3. FOR SMALL NFC ANTENNAS, EXTERNAL CRYSTAL OSCILLATOR CIRCUIT MUST BE USED - SEE EE8994CORE-296 FOR DETAILS. IF USE OF EXTERNAL CLOCK IS DESIRED, CHECK WITH BROADCOM BEFORE COMMITTING.

APPROVALS		 MOTOROLA MOBILE DEVICES	
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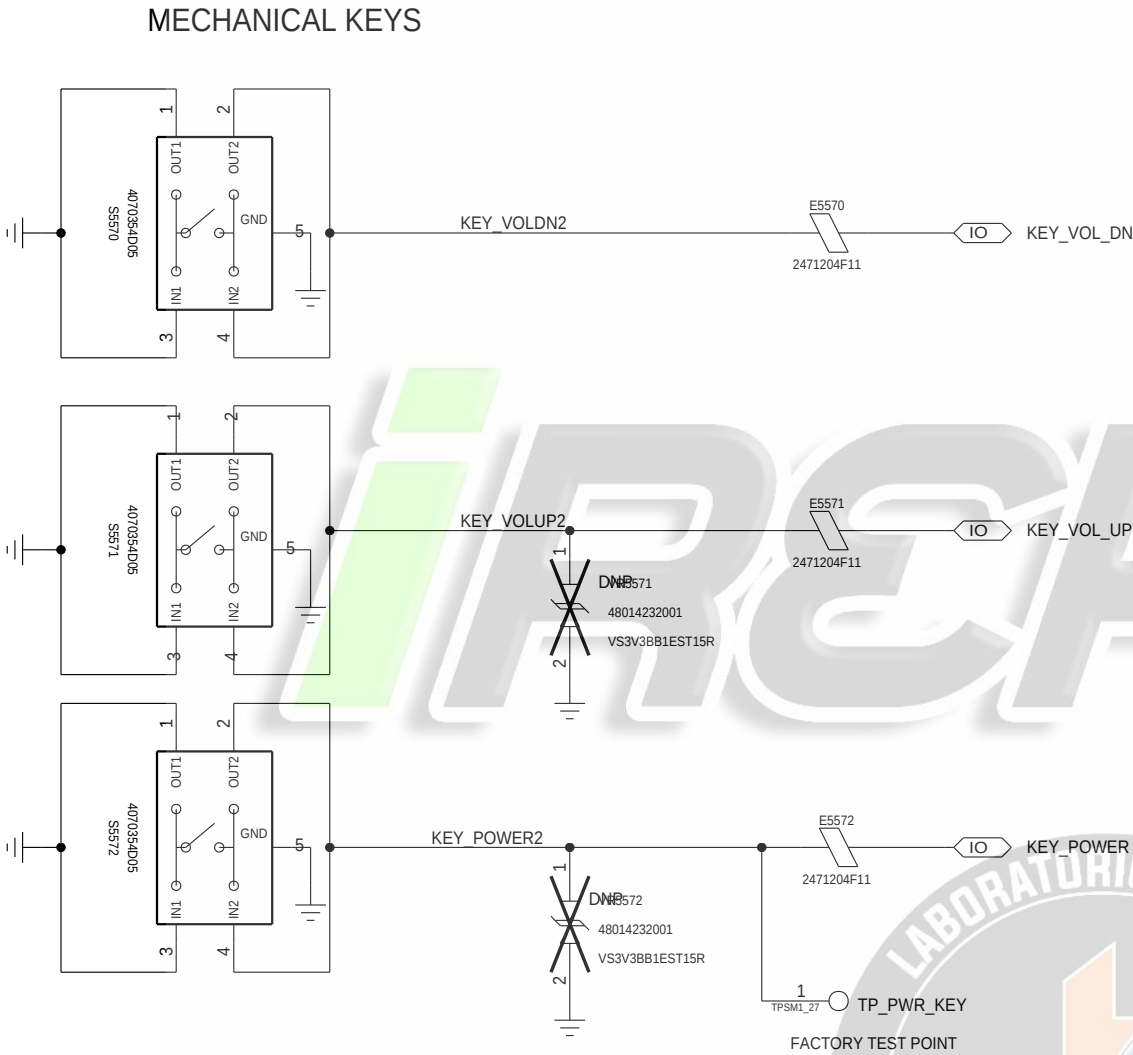
USR: FPC



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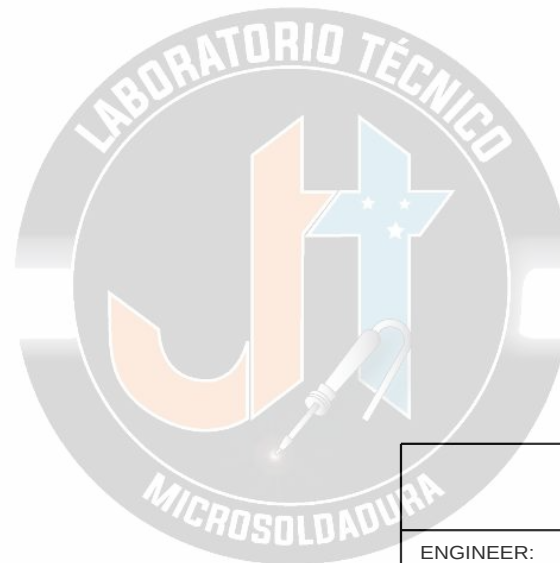
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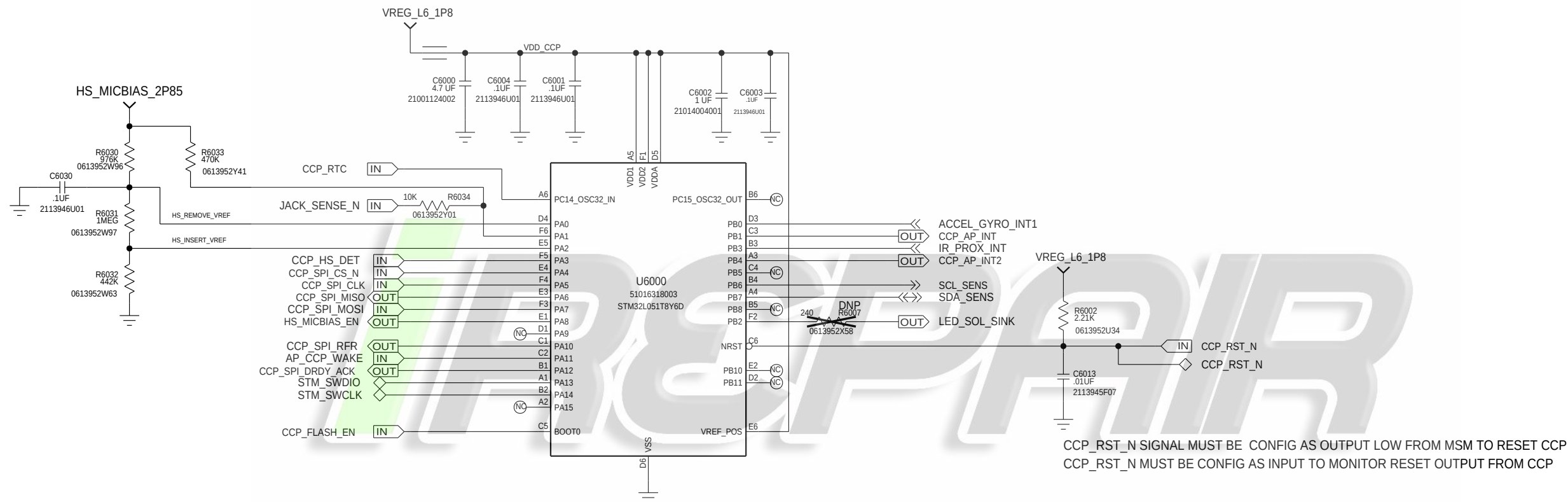
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REF:5800-5819

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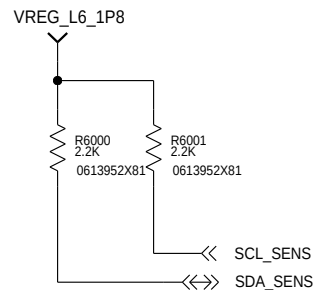
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REF: 6000-6049

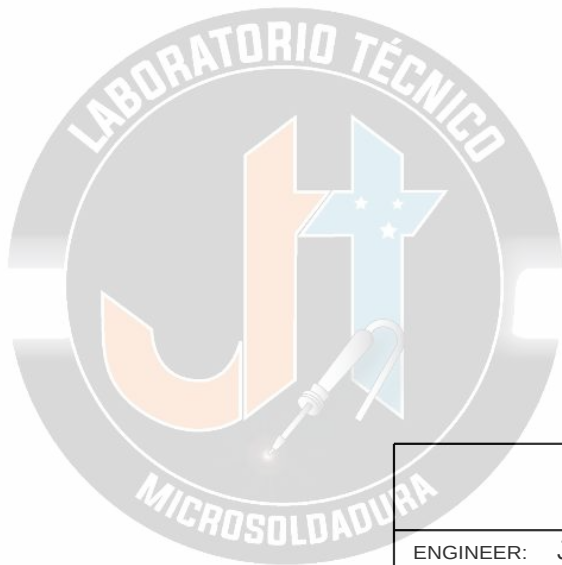
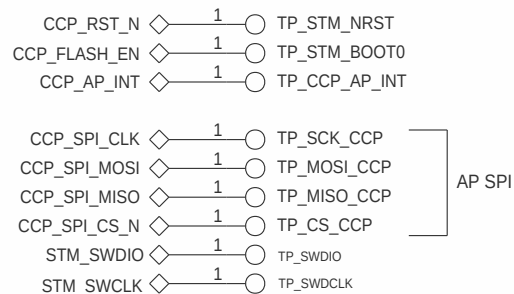


CCP_RST_N SIGNAL MUST BE CONFIG AS OUTPUT LOW FROM MSM TO RESET CCP
CCP_RST_N MUST BE CONFIG AS INPUT TO MONITOR RESET OUTPUT FROM CCP

I2C SUPPORT



TESTPOINTS



APPROVALS

ENGINEER: JOHN GORSICA

LAST MODIFIED BY: NATE CONNELL
Fri Feb 19 14:26:16 2016

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BLOCK TITLE:

xmcs

PROJECT NAME
84016530001_a.cpm

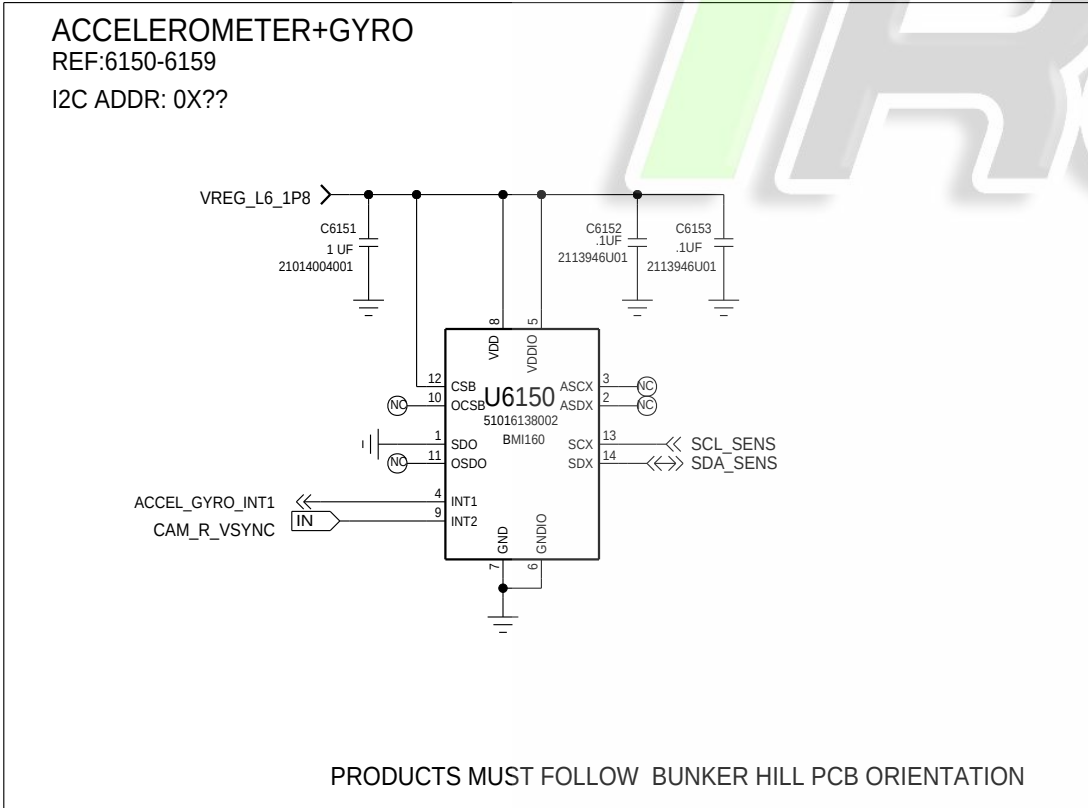
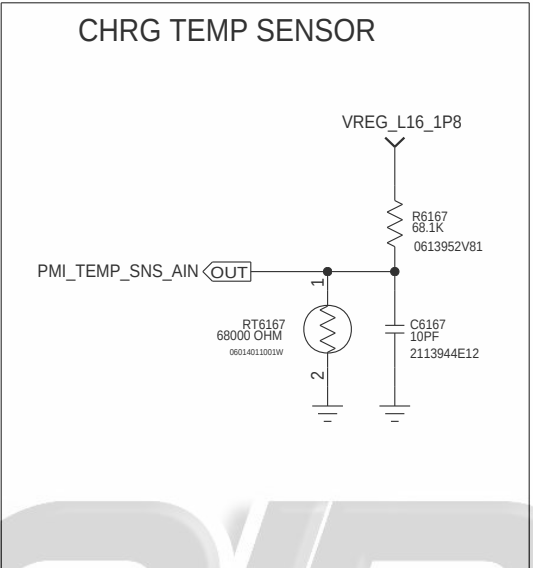
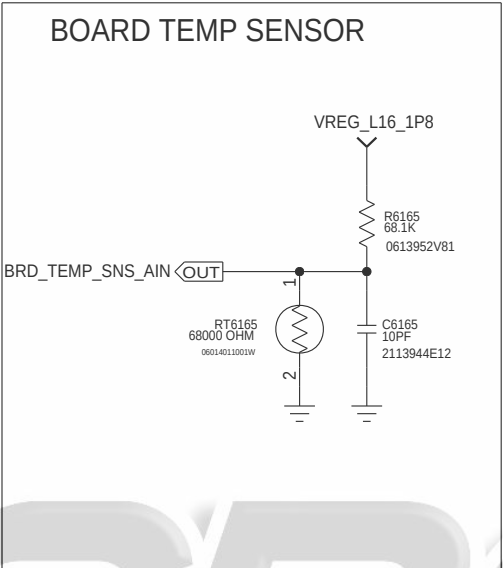
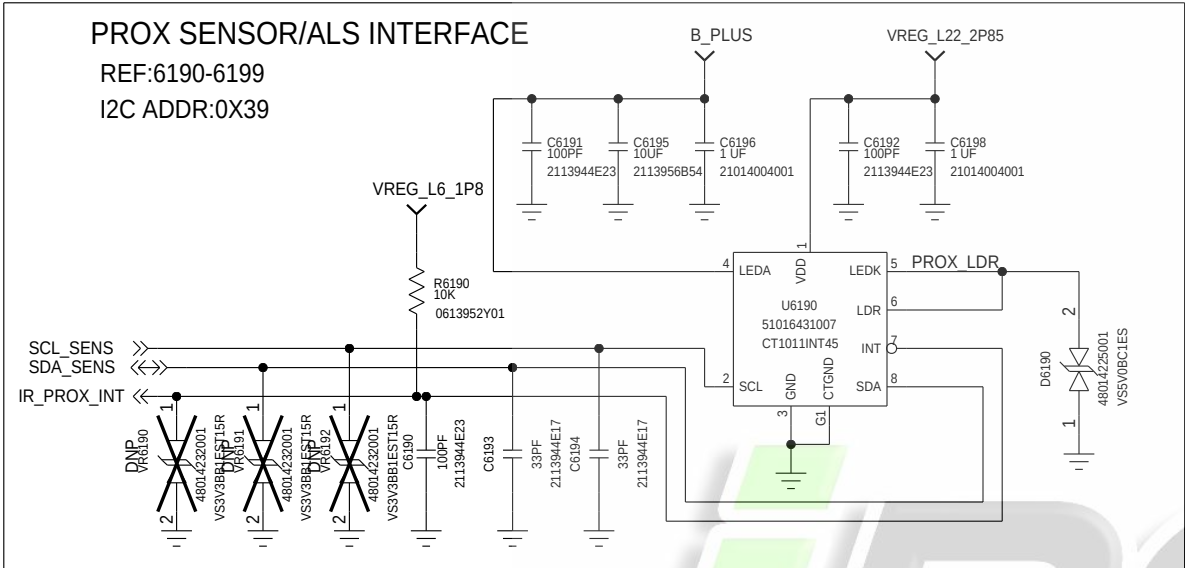
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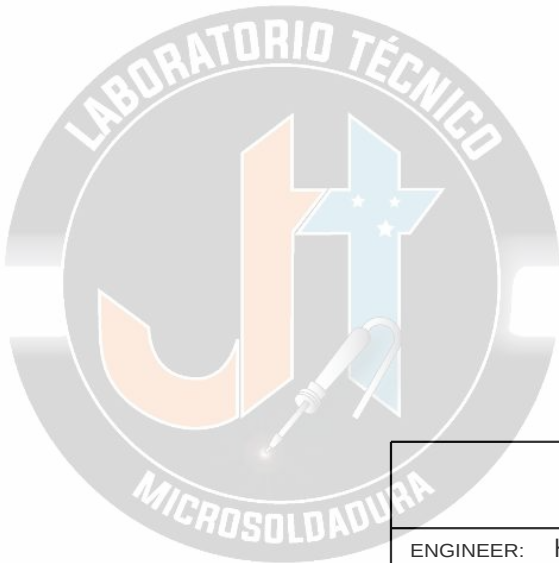
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XMCS: SENSORS

REF: 6100-6199



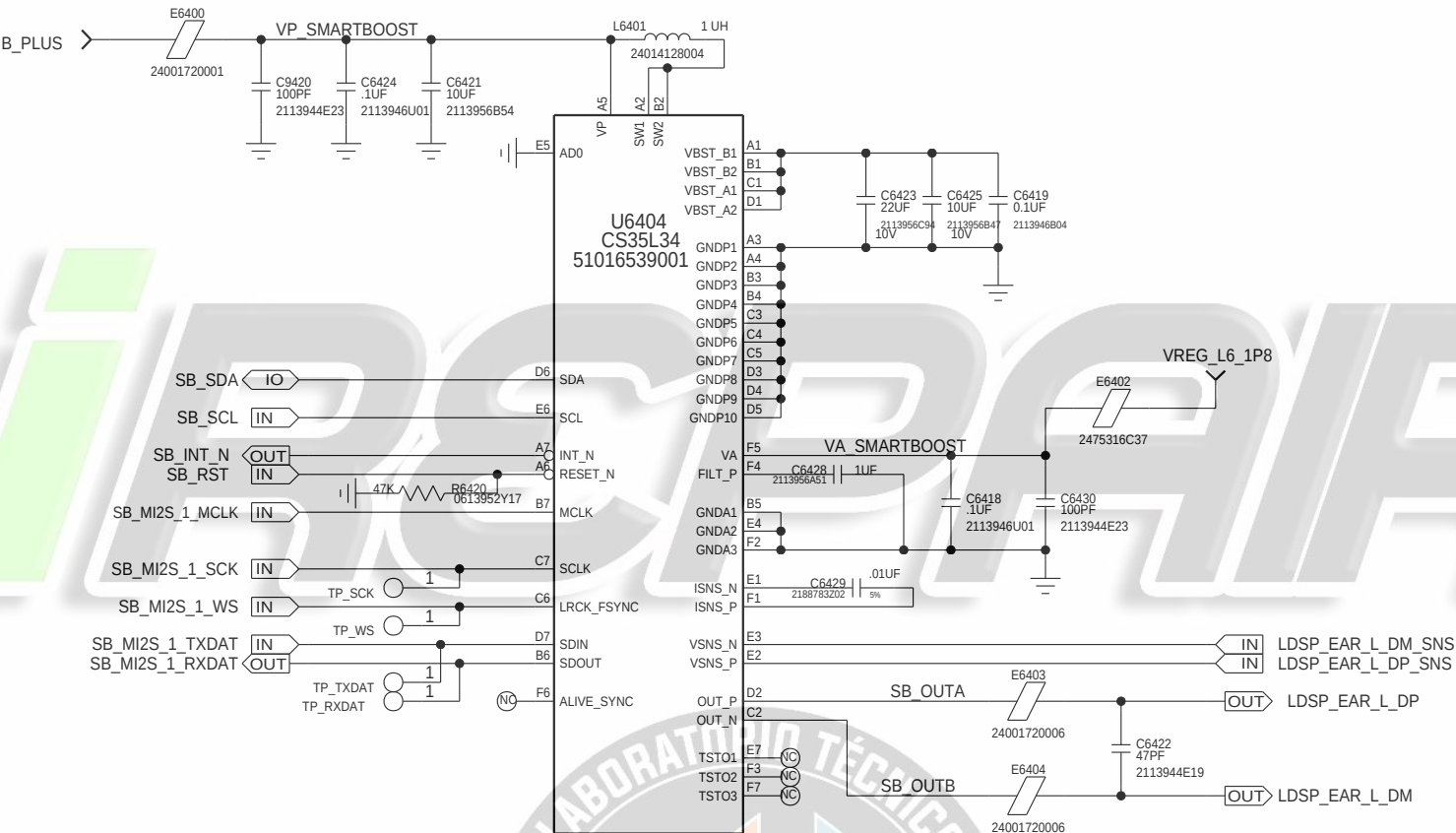
PRODUCTS MUST FOLLOW BUNKER HILL PCB ORIENTATION



JOHN GORSICA/NATE CONNELL APPROVALS		 MOTOROLA MOBILE DEVICES	
ENGINEER: HUNG PHAM		BLOCK TITLE:	
LAST MODIFIED BY: HUNG PHAM Fri Feb 19 14:26:17 2016		xmcs	
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		CAD FILE NAME:	
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		PAGE 18 OF 52	

XMCS: SMART BOOST WATTSON

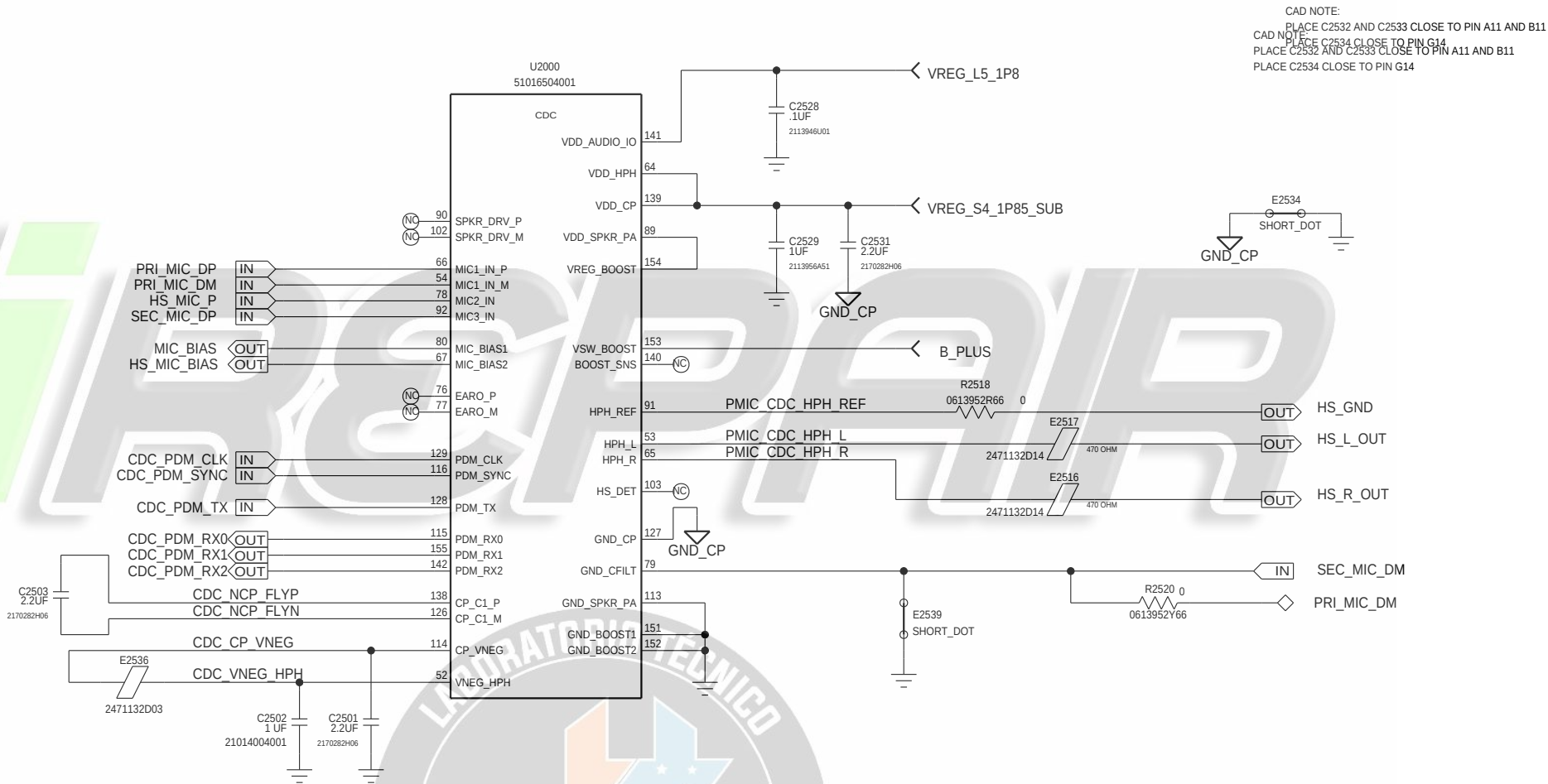
REF 6400-6499



APPROVALS		 MOTOROLA MOBILE DEVICES	
ENGINEER:		BLOCK TITLE:	
LAST MODIFIED BY:		xmcs	
Fri Feb 19 14:26:17 2016			
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		CAD FILE NAME:	
		PAGE 19 OF 52	

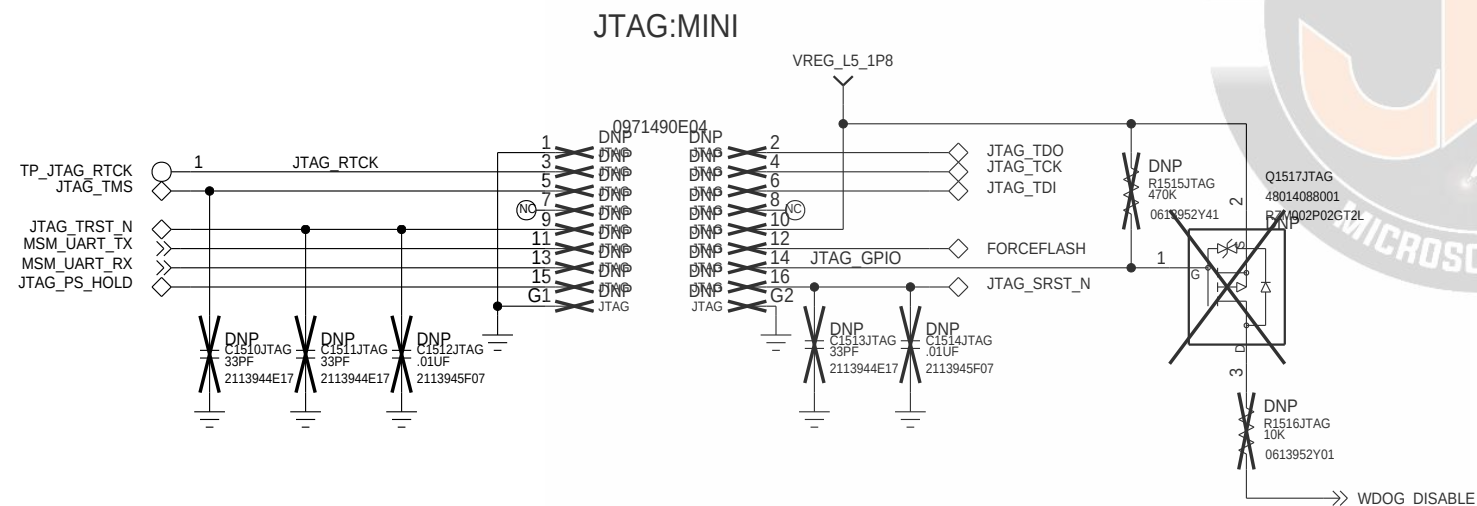
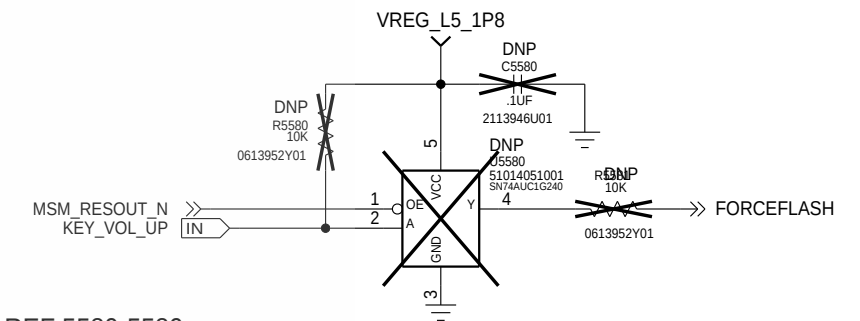
XMCS: CODEC

REF 2500-2579



APPROVALS		 MOTOROLA MOBILE DEVICES		
ENGINEER:		BLOCK TITLE: xmcs		
LAST MODIFIED BY: FAN021 Fri Feb 19 14:26:16 2016				
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		B 84016530001_a.cpm		01
		CAD FILE NAME:		PAGE 20 OF 52

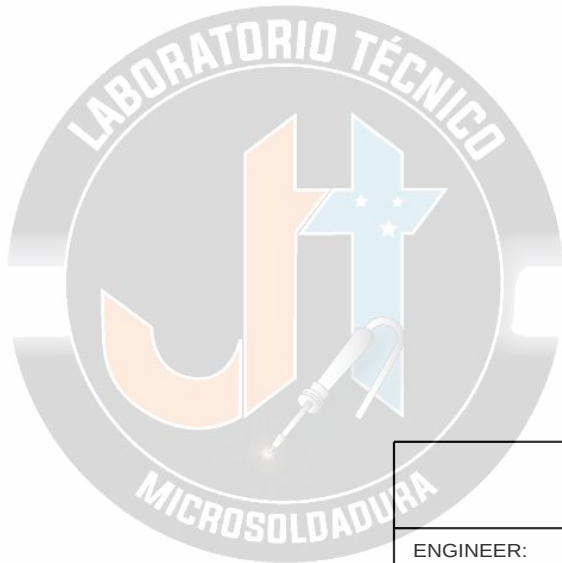
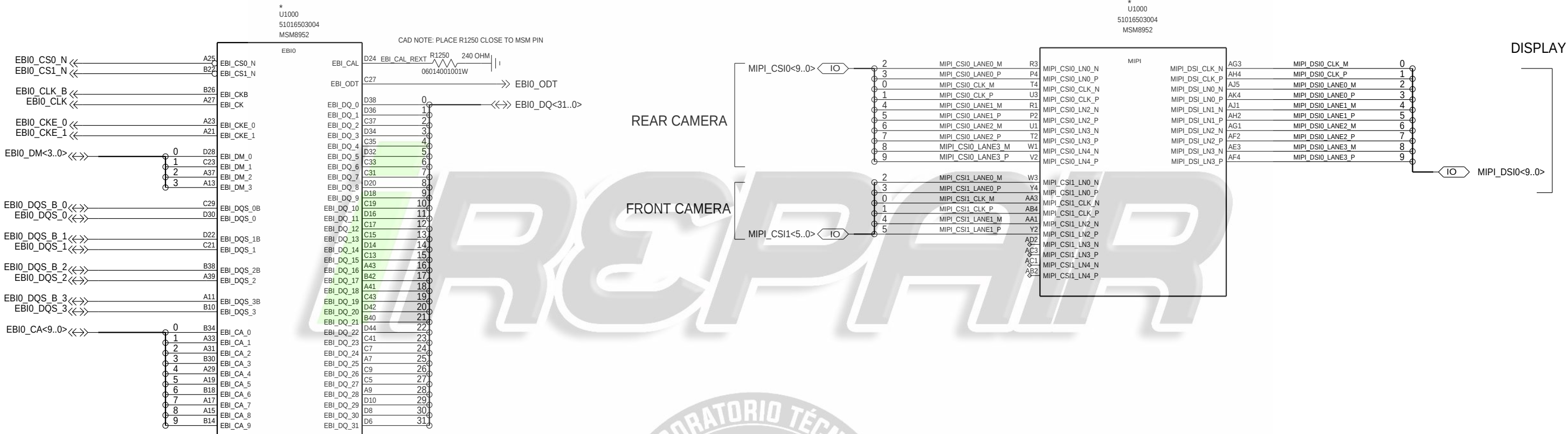
REF 1000-1099



APPROVALS		 MOTOROLA MOBILE DEVICES	
ENGINEER:	BLOCK TITLE: ap_pmic		
LAST MODIFIED BY: FAN021 Fri Feb 19 14:26:18 2016			
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		CAD FILE NAME:	PAGE 21 OF 52

MSM: MIPI/DSI/CSI/EBIO

REF 1200-1299



APPROVALS		MOTOROLA MOBILE DEVICES	
ENGINEER:		BLOCK TITLE:	
LAST MODIFIED BY: Fri Feb 19 14:26:26 2016		ap_pmic	
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		CAD FILE NAME:	
		PAGE 22 OF 52	

MSM: GPIO (0-133)

REF 1200-1299

NOTE: RESERVE GPIO_3 FOR WDOG_DISABLE FUNCTION.
DO NOT HAVE A PULL-UP ON THIS GPIO FOR BETTER
SW DEBUG CAPABILITY.

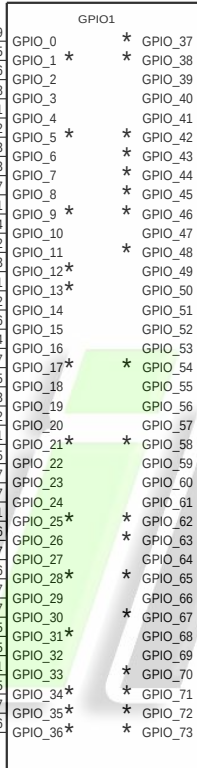
U1000
MSM8952
51016503004

NOTE: ASTERISK(*) INDICATE MODEM POWER MANAGEMENT (MPM) WAKE-UP PINS.

NOTE: DO NOT HAVE PULL-UPS ON THE FOLLOWING GPIOs UNLESS INTEND FOR BOOT
OR SECURE BOOT RELATED CONFIGURATIONS:
GPIO_8, 9, 91, 107, 108, 109

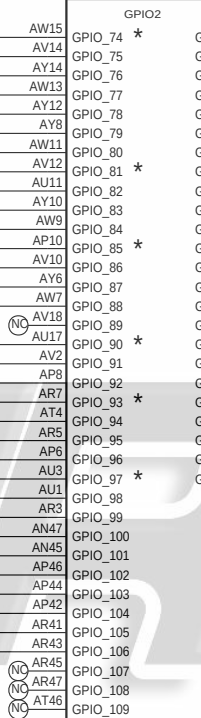
U1000
MSM8952
51016503004

DISP_RST_N
CCP_SPI_DRDY_ACK
WDG_DISABLE
MSM_UART_TX
MSM_UART_RX
BLSP2_I2C_SDA
BLSP2_I2C_SCL
CCP_FLASH_EN
CCP_AP_INT
TS_I2C_SDA
TS_I2C_SCL
CCP_SPI_RFR
NFC_IRQ_OUT
NFC_I2C4_SDA
NFC_I2C4_SCL
CCP_SPI_MOSI
CCP_SPI_MISO
CCP_SPI_CS_N
CCP_SPI_CLK
DTV_SPI_MOSI
DTV_SPI_MISO
DTV_SPI_CS
DTV_SPI_CLK
DISP_TE
SB_MI2S_1_MCLK
CAM_R_MCLK
CAM_F_MCLK
CCI_I2C_SDA
CCI_I2C_SCL
DISP_1P8_ENBL
CAM_R_RST_N



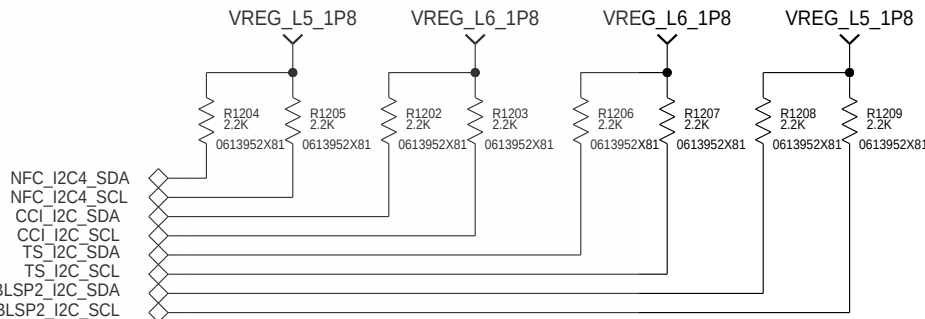
DTV_INT_N
DTV_RST_N
CAM_F_RST_N
DTV_1P2_EN
FP_IRQ_N
FOCUS_GPIO1
PA_ON
FP_RST_N
UIM_BATT_ALARM
FORCEFLASH
UIM1_DATA
UIM1_CLK
UIM1_RESET
UIM2_DATA
UIM2_CLK
UIM2_RESET
UIM2_PRESENT
SAR_CUTBACK
USB_HS_ID
SB_INT_N
TS_RST_N
TS_INT_N
HAP_PWM
SD_CARD_DET_N
EAR_SPKR_SEL
CDC_PDM_CLK
CDC_PDM_SYNC
CDC_PDM_TX
CDC_PDM_RX0
CDC_PDM_RX1

CDC_PDM_RX2_IN
BT_SSBK
WLAN_DATA<2.0>
WLAN_SET_OUT
WLAN_CLK_OUT
FM_SSBK
FM_DATA
BT_CTL_OUT
BT_DATA
KEY_VOL_UP
FOCUS_XSHUT_OUT
CCP_RESOUT_OUT
SB_MI2S_TXDAT_OUT
CCP_AP_INT2_IN
SB_MI2S_1_SCK_OUT
SB_MI2S_1_WS_OUT
SB_MI2S_1_RXDAT_IN
SB_RST_OUT
RF_CONN_DET_DIV
FP_SPI_MOSI
FP_SPI_MISO
FP_SPI_CS_N
FP_SPI_CLK
DRX_CA_CTL2_OUT
RF_CONN_DET_LB_HB_IN
CTL_DIV_SPDT_OUT
CTL_PRX_SPDT_OUT
CTL2_PRX_SPDT_OUT
CTL_PDET_SPDT_OUT
CTL_DA_SPDT_OUT



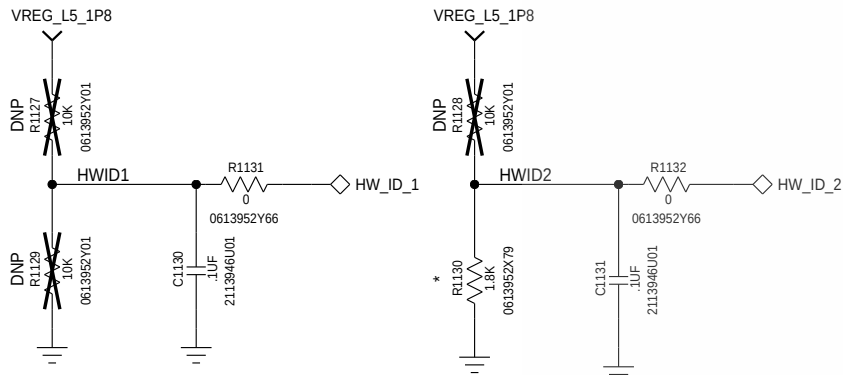
DRX_CA_CTL
SAR_CUTBACK
CAM_LDO_ENBL
PA_ON
EXT_GPS_LNA_EN
CH0_GSM_TX_PHASE_D0
RFFE1_CLK
RFFE1_DATA
RFFE2_CLK
RFFE2_DATA
RFFE4_CLK
RFFE4_DATA
RFFE5_CLK
RFFE5_DATA
AP_CCP_WAKE
NFC_PWR_ON
NFC_ENABLE_IN
HW_ID_1
HW_ID_2

I2C PULL-UP RESISTORS



MOT BOARD ID

FOR RUNTIME REVISION INFORMATION, REFER TO:
<https://sites.google.com/a/motorola.com/runtime-product-ids>



WAITING FOR GPIO ASSIGNMENT
NEED WAKABLE

APPROVALS

ENGINEER:

LAST MODIFIED BY: FAN021
Fri Feb 19 14:26:28 2016

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MOTOROLA
MOBILE DEVICES

BLOCK TITLE:

ap_pmic

PROJECT NAME

84016530001_a.cpm

REV

01

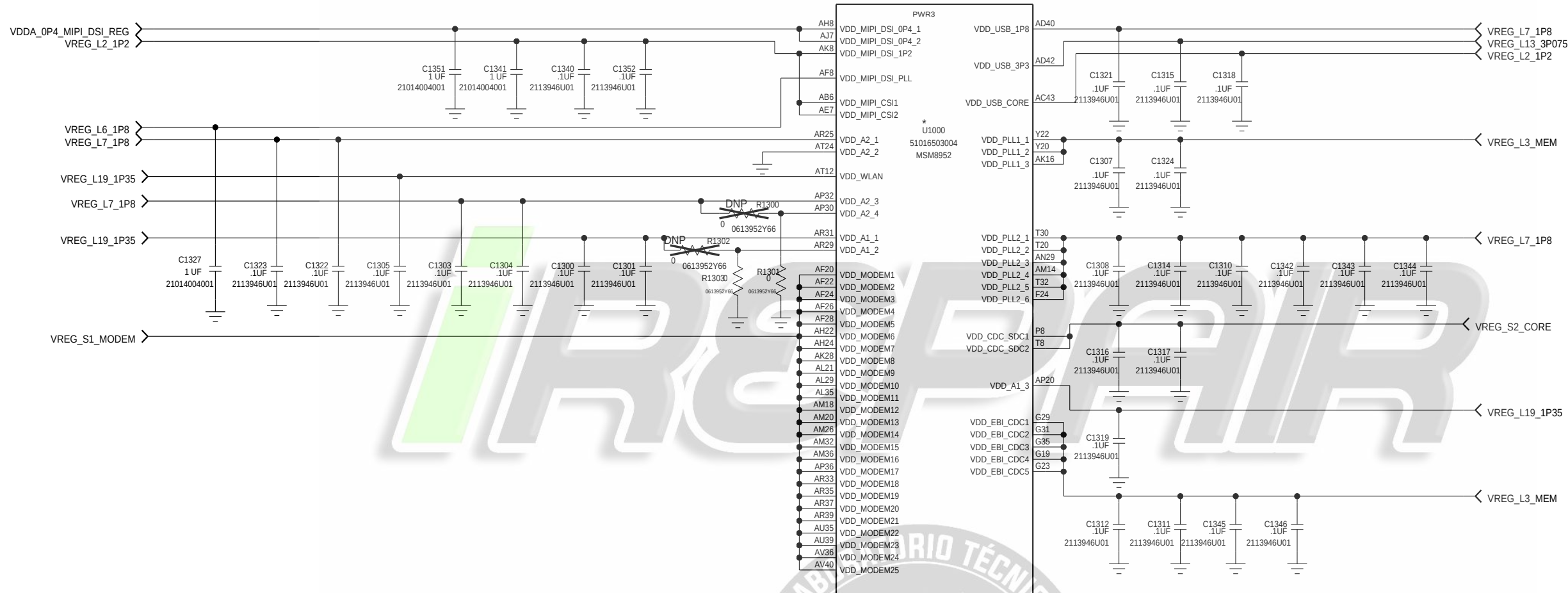
CAD FILE NAME:

PAGE 23 OF 52

MSM: PWR1

REF 1300-1399

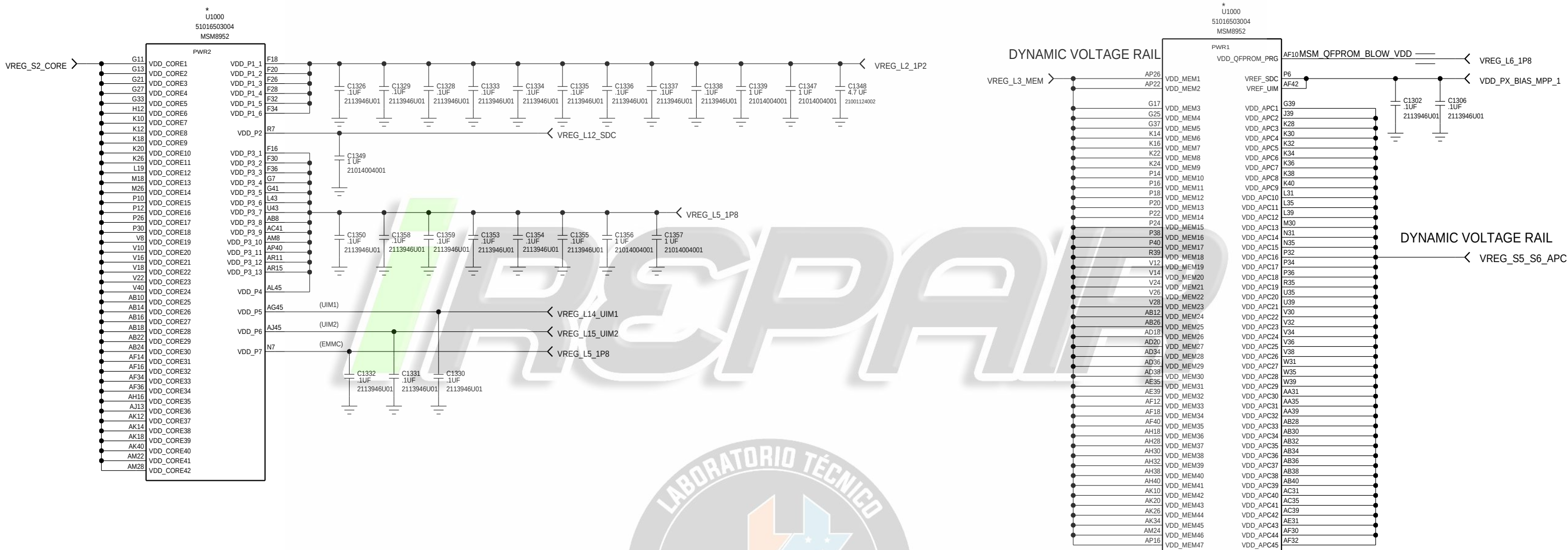
CAD NOTE: PLACE ONE DECAP NEAR PINS V5, V6 AND
THE OTHER DECAP NEAR PIN R5



APPROVALS		 MOTOROLA MOBILE DEVICES	
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		REV 01	
		CAD FILE NAME:	
		PAGE 24 OF 52	

MSM: PWR2

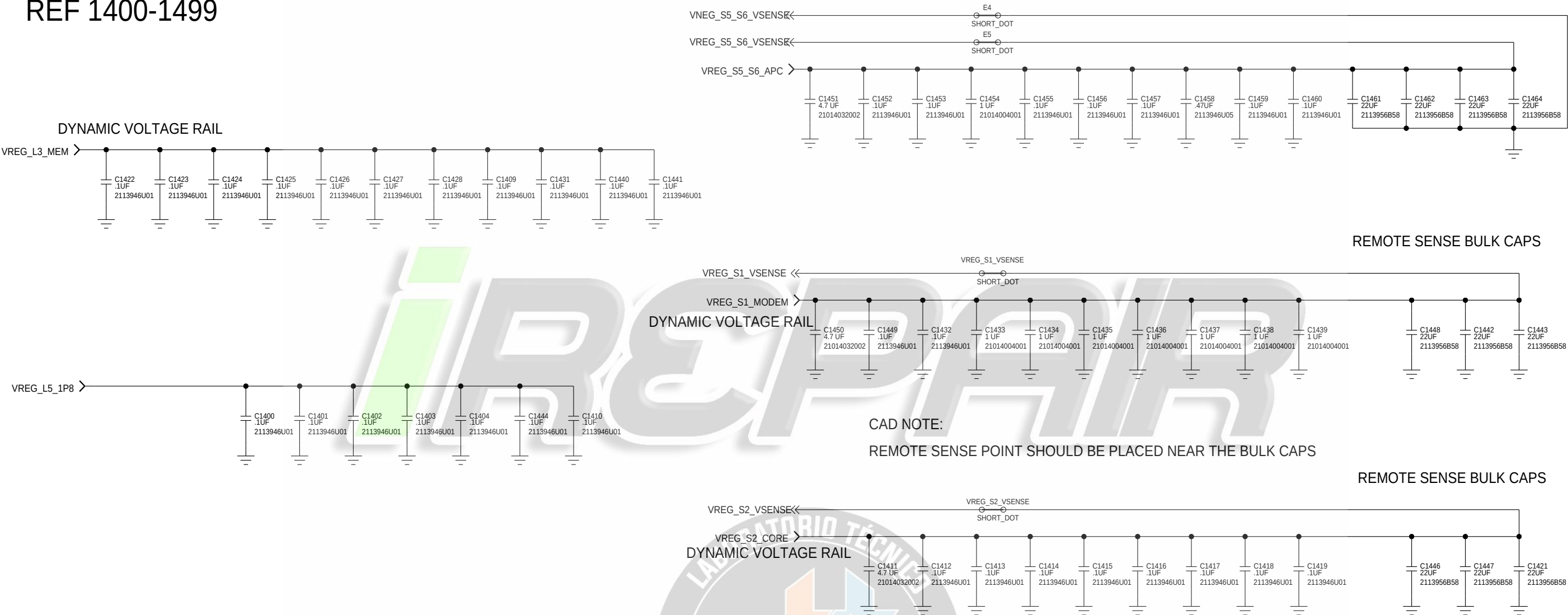
REF 1300-1399



APPROVALS		 MOTOROLA MOBILE DEVICES	
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LAST MODIFIED BY: FAN021 Fri Feb 19 14:26:29 2016		ap_pmic	
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		CAD FILE NAME:	PAGE 25 OF 52

MSM: BYPASS CAPS

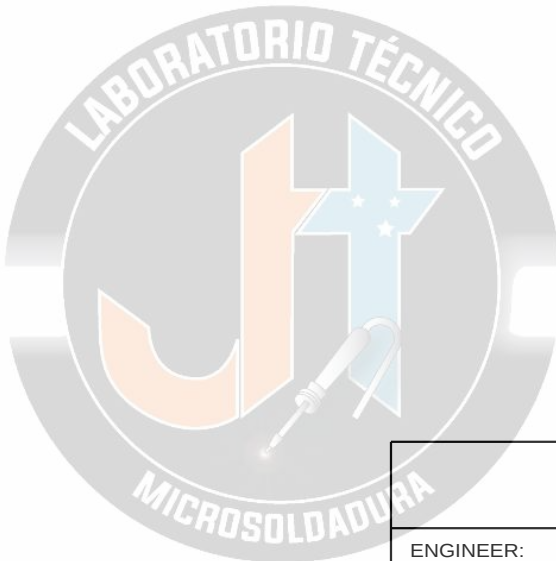
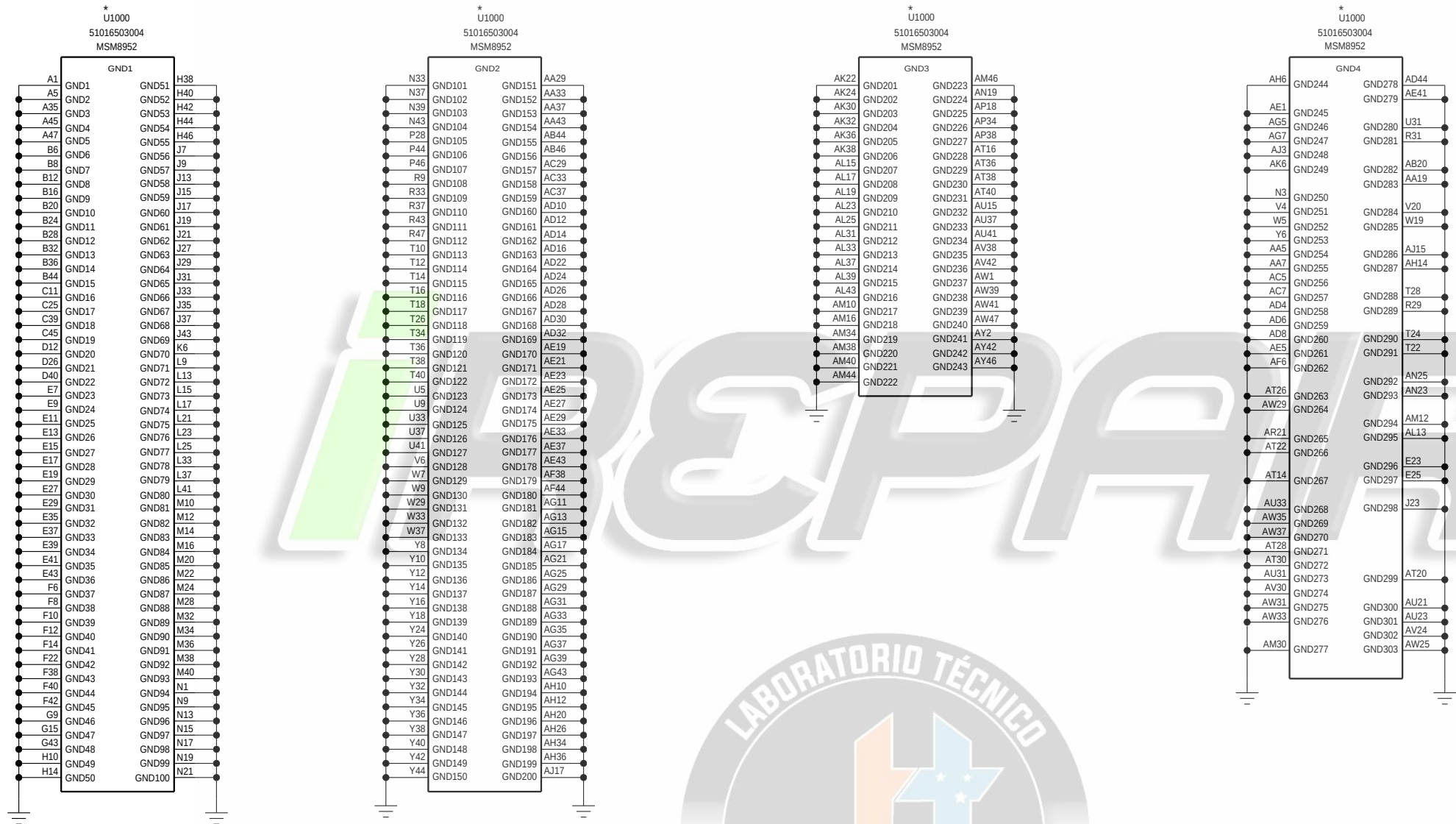
REF 1400-1499



CAD NOTE:
REMOTE SENSE POINT SHOULD BE PLACED NEAR THE BULK CAPS

APPROVALS		MOTOROLA MOBILE DEVICES	
ENGINEER:		BLOCK TITLE:	
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		CAD FILE NAME:	PAGE 26 OF 52

MSM: GROUND



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		B 84016530001_a.cpm	REV 01
		CAD FILE NAME:	
		PAGE 27 OF 52	

MSM: MCP MEMORY

MCP: 8GB NAND, 8GBIT LPDDR3

D

D

C

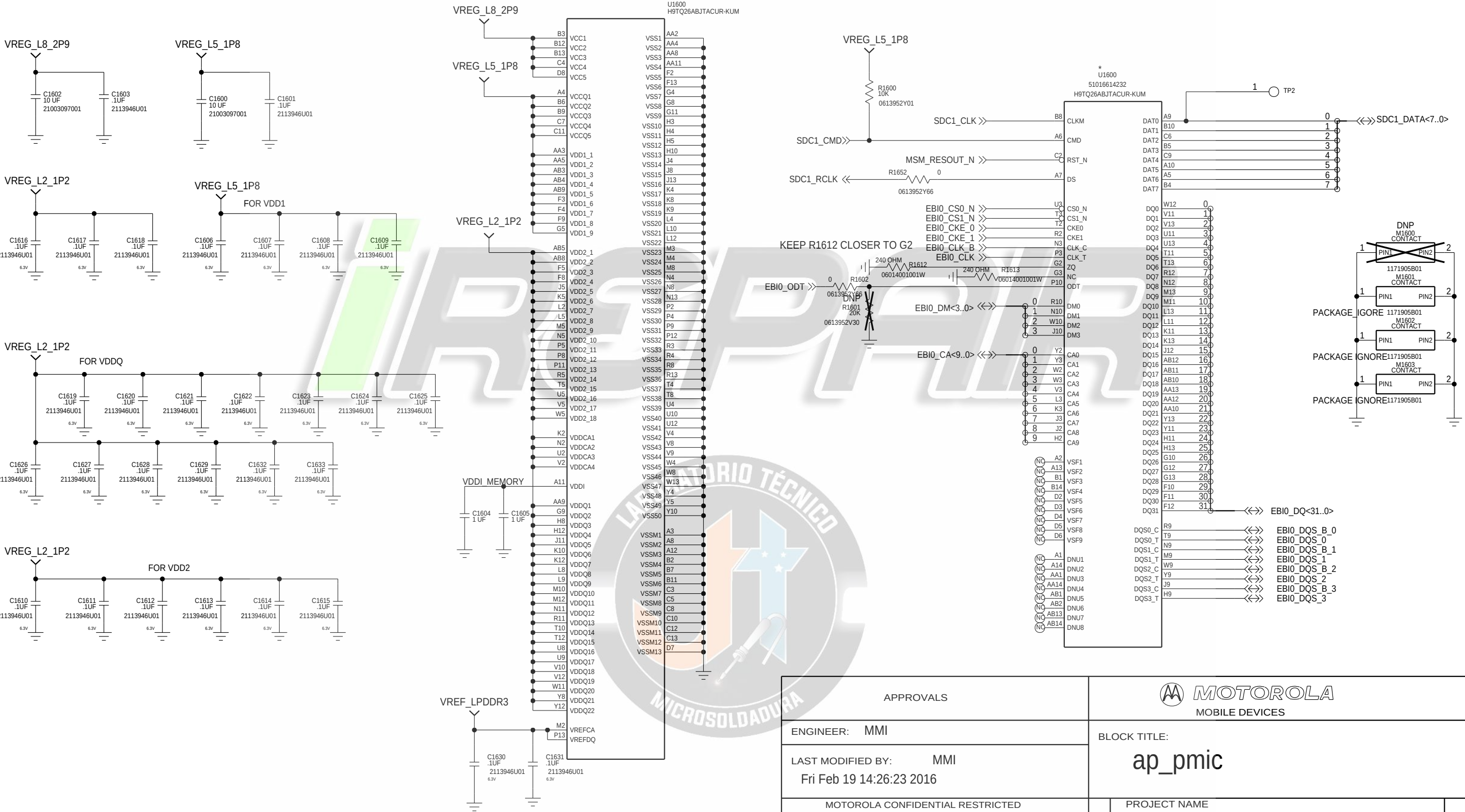
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B

B

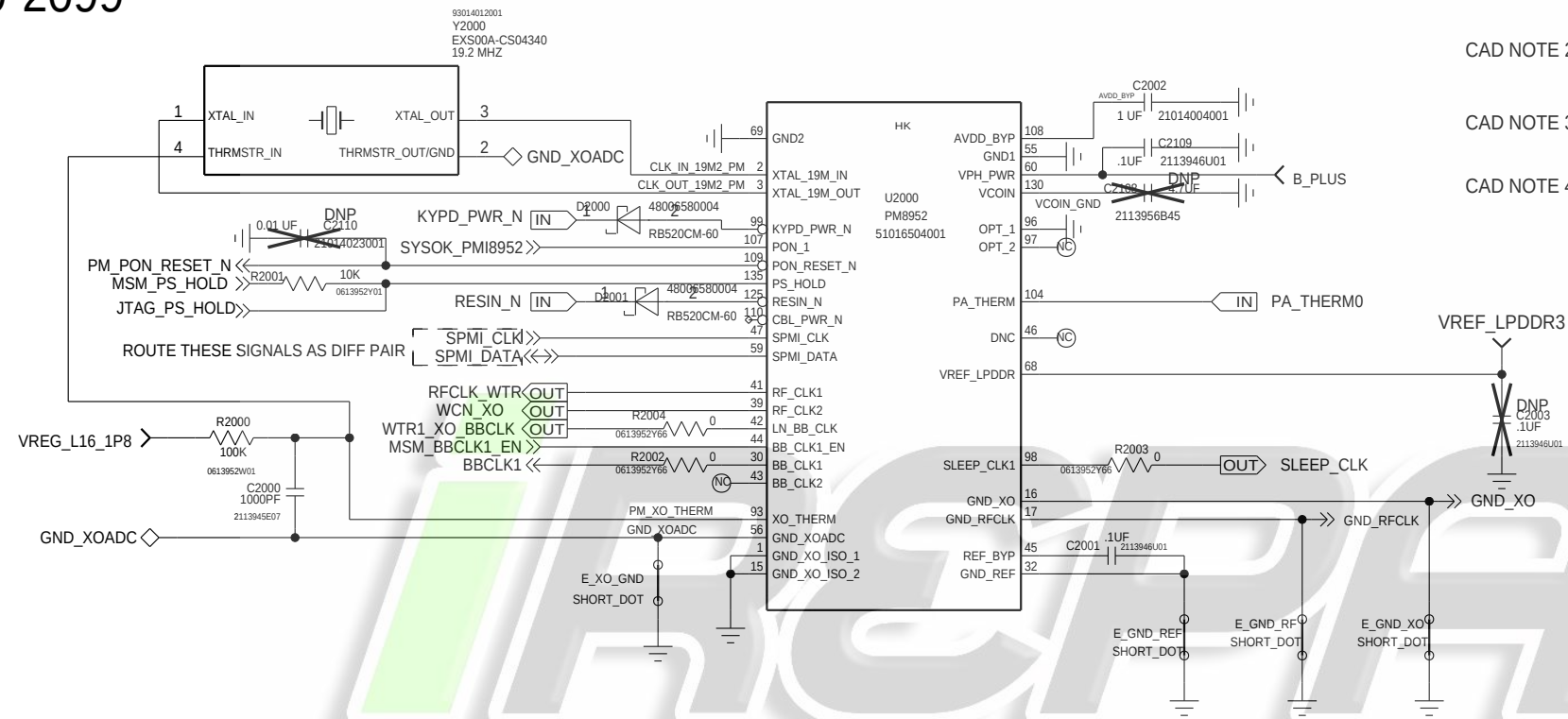
A

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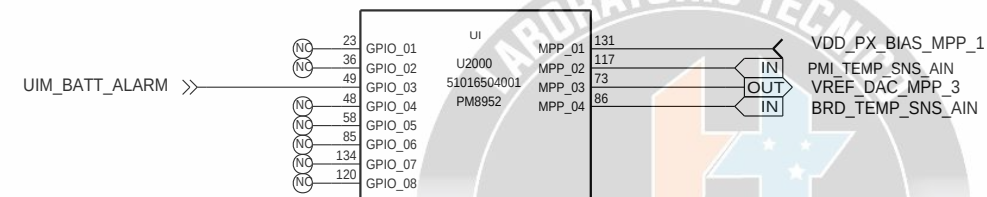


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ENGINEER: MMI		BLOCK TITLE:	
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		PAGE 28 OF 52	

REF: 2000-2099



NOTE: CONNECT GND_REG AND REF_BYP TO C2001 ON TOP LAYER WITHOUT VIA



CAD NOTE 1: PLACE C2000 CLOSE TO Y5 AND CONNECT CAP GND THROUGH DEDICATED VIA TO MAIN GND PLANE

CAD NOTE 2: CONNECT J3 (GND_RF) TO NEGATIVE TERMINAL OF VREG_RFCLK AND CONNECT TO THE MAIN GND USING DIRECT VIAS

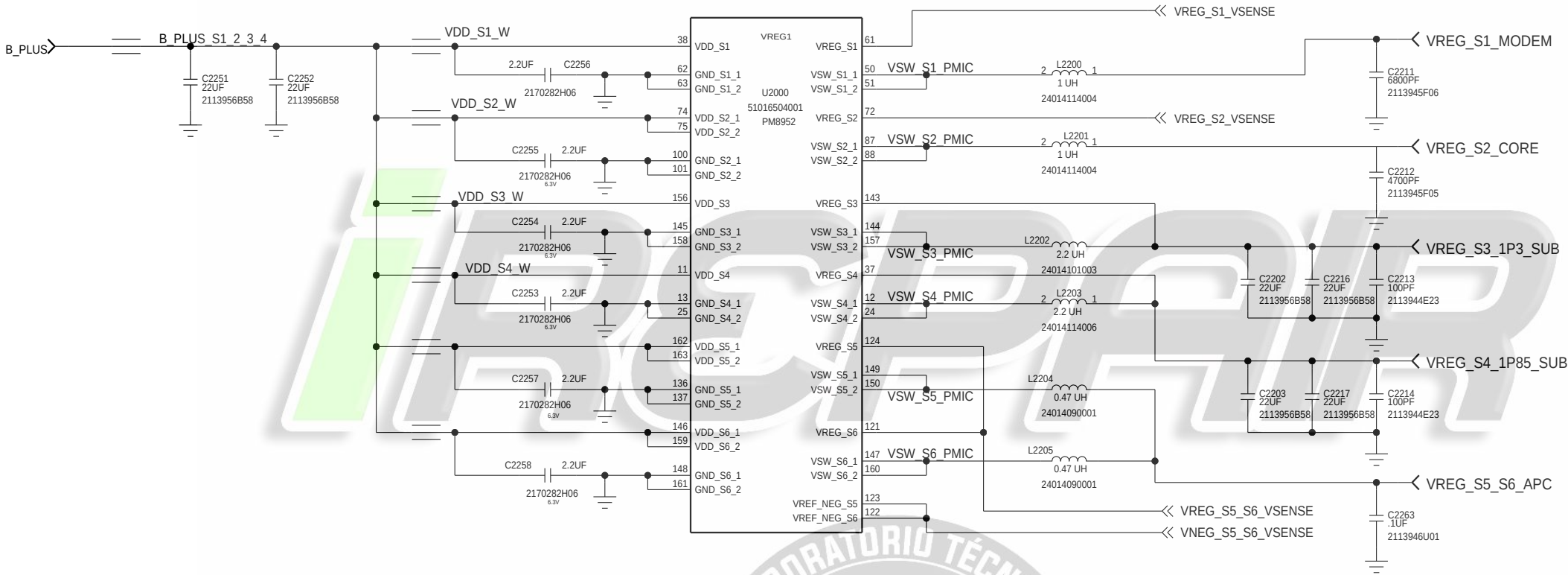
CAD NOTE 3: CONNECT J2 (GND_XO) TO NEGATIVE TERMINAL OF VREG_XO AND CONNECT TO THE MAIN GND USING DIRECT VIAS

CAD NOTE 4: H3 MUST BE CONNECTED TO MAIN GND USING DEDICATED VIA

APPROVALS		 MOTOROLA MOBILE DEVICES	
ENGINEER: MMI	BLOCK TITLE: ap_pmic		
LAST MODIFIED BY: MMI Fri Feb 19 14:26:20 2016			
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		CAD FILE NAME:	PAGE 29 OF 52

PM8952: SMPS

REF: 2200-2299

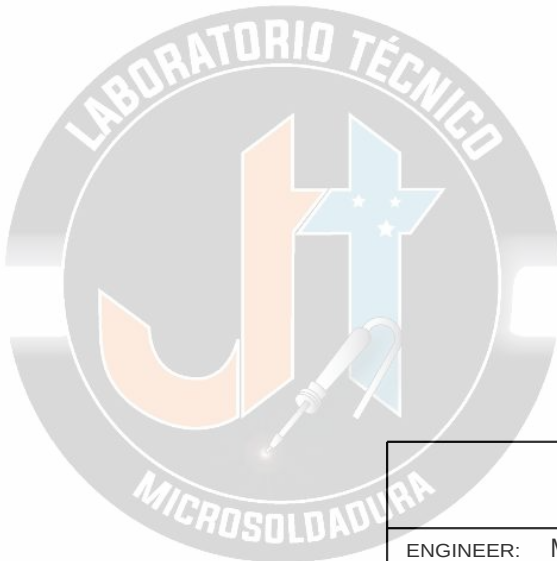
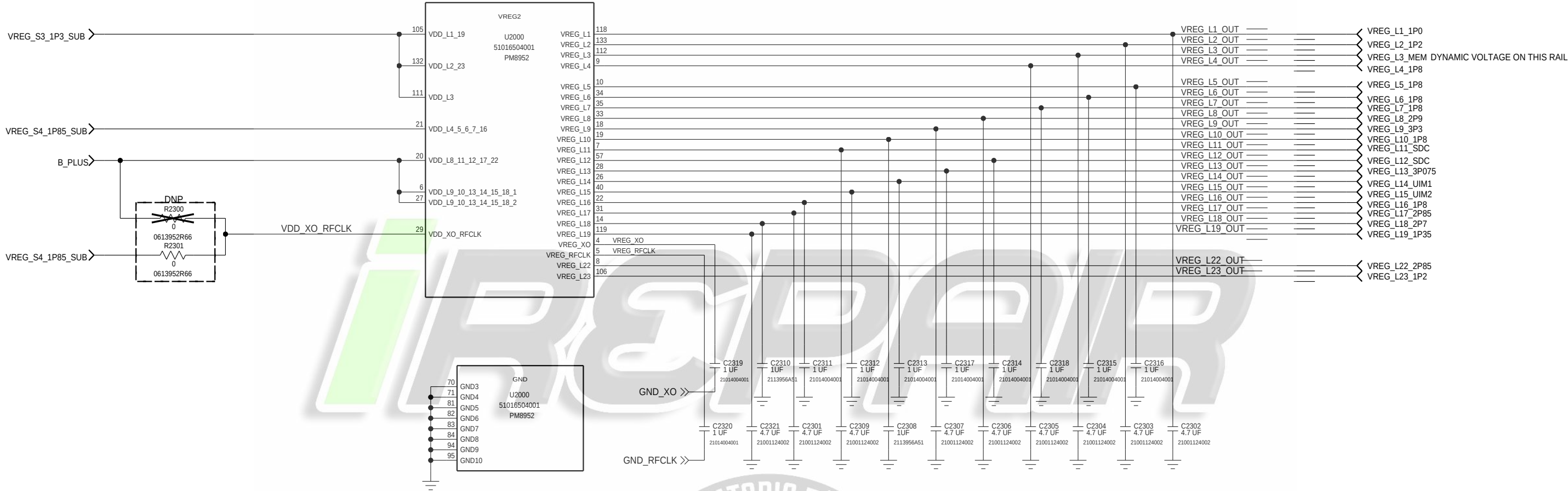


NOTE: PLACE INPUT CAPS CLOSE TO RESPECTIVE VIN PINS
EACH CAP GND SHOULD BE STAR-ROUTED

APPROVALS		 MOTOROLA MOBILE DEVICES	
ENGINEER: MMI		BLOCK TITLE: ap_pmic	
LAST MODIFIED BY: MMI Fri Feb 19 14:26:25 2016		PROJECT NAME 84016530001_a.cpm	
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		PAGE 30 OF 52	

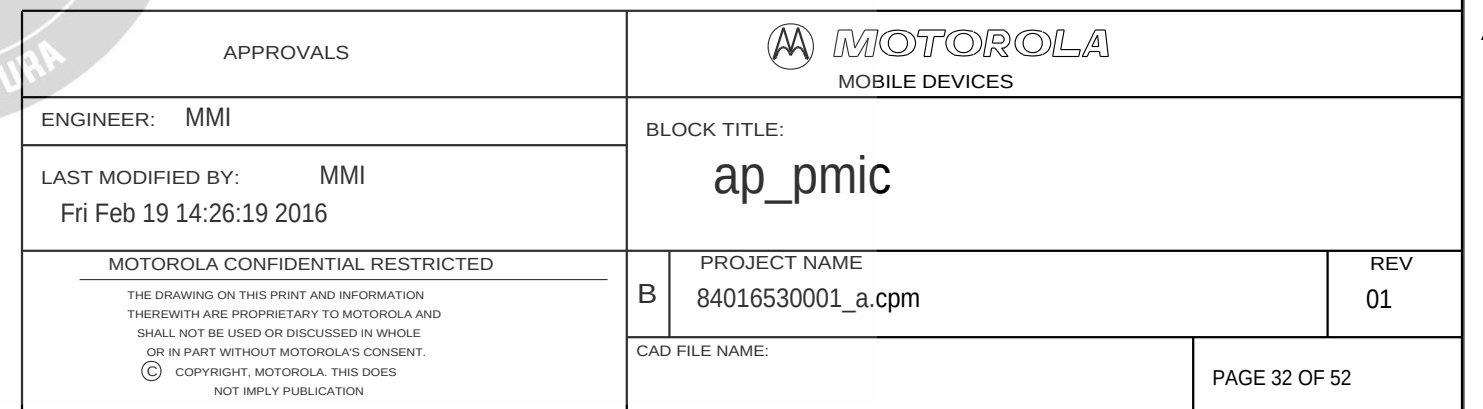
PM8952: VREGS

REF: 2300-2399



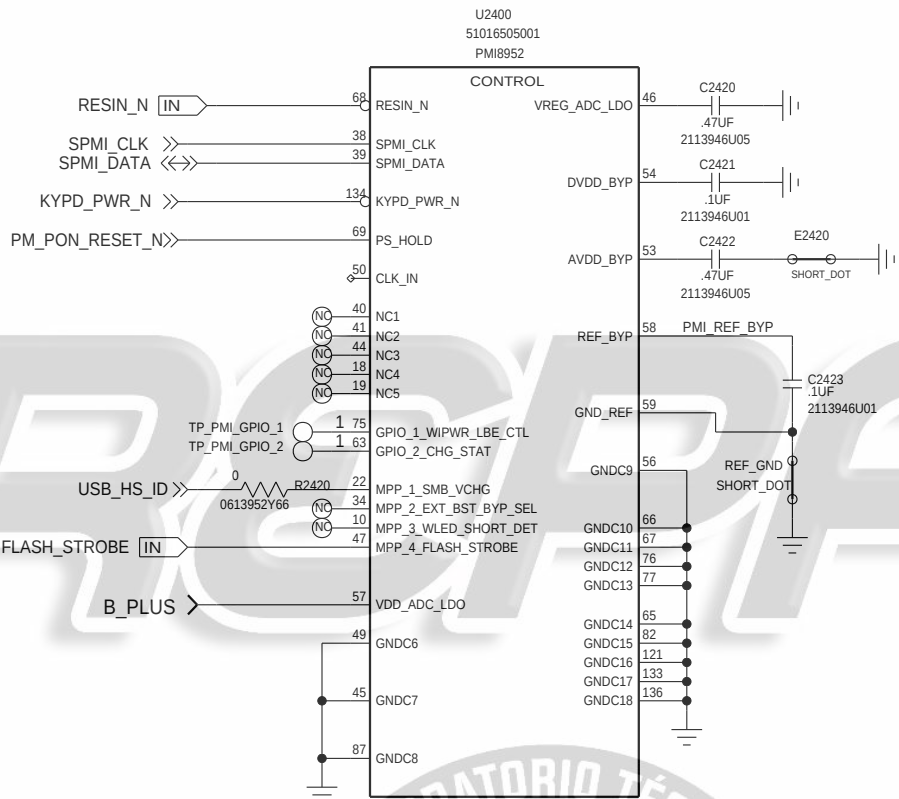
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		B 84016530001_a.cpm	01
		CAD FILE NAME:	PAGE 31 OF 52

REF: 2400-2419



PMI8952: IO_REF

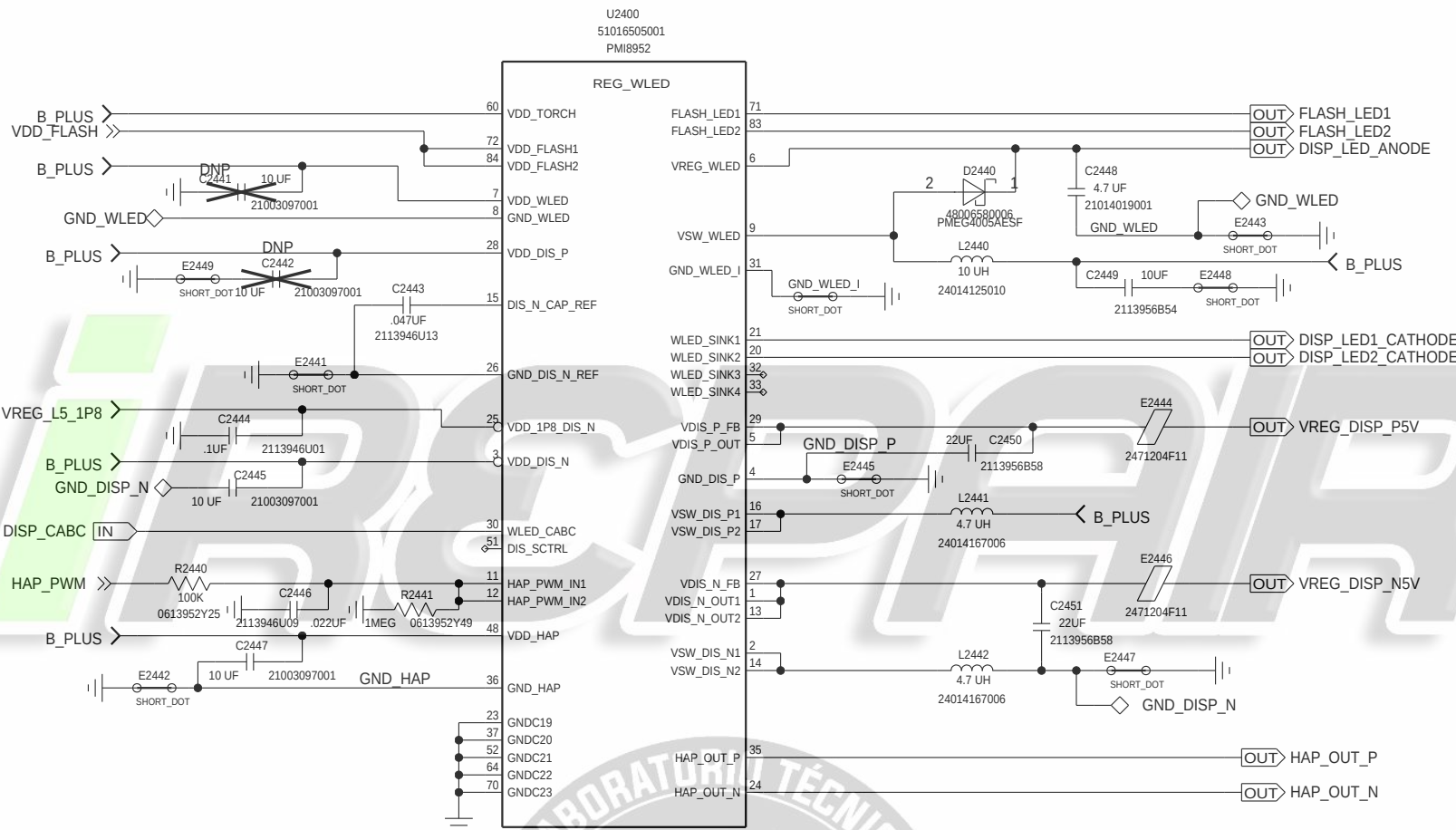
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APPROVALS		 MOTOROLA MOBILE DEVICES	
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		CAD FILE NAME:	
		PAGE 33 OF 52	

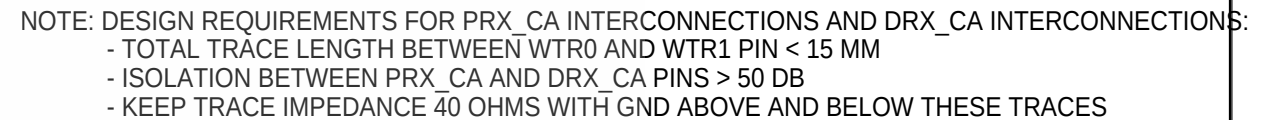
PMI8952: REG_WLED

REF: 2440-2459



APPROVALS		 MOTOROLA MOBILE DEVICES	
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LAST MODIFIED BY: MMI Fri Feb 19 14:26:21 2016		ap_pmic	
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REF 3000-3049

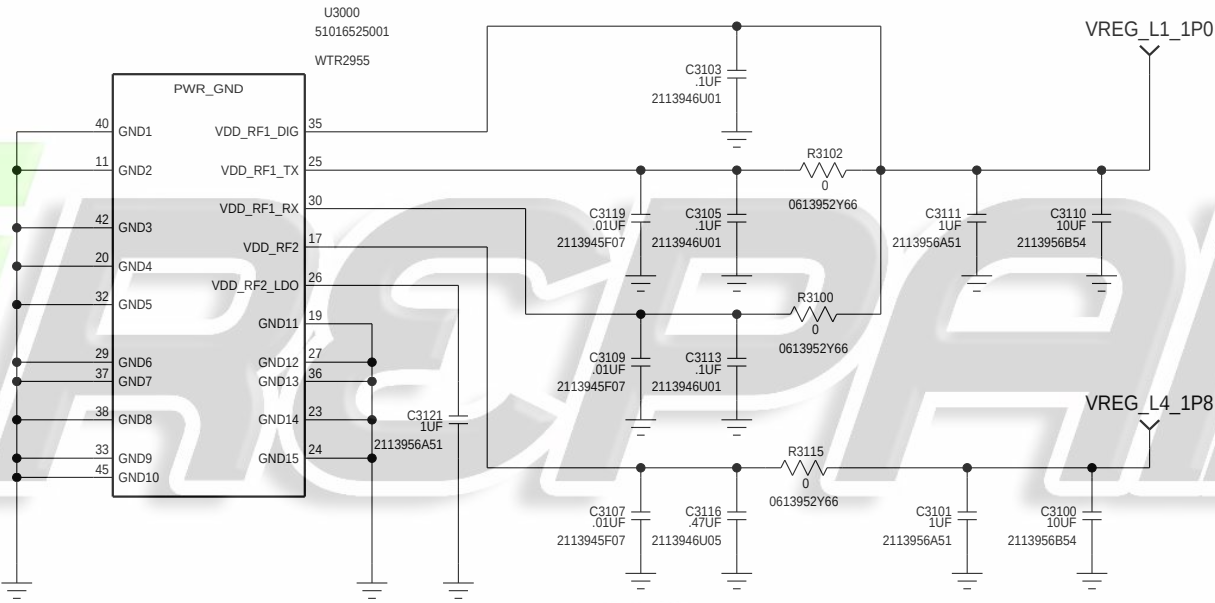


MOTOROLA MOBILITY CONFIDENTIAL RESTRICTED-MDB TRACKED TRACKEDANDBLOCKED

APPROVALS		 MOTOROLA MOBILE DEVICES	
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		CAD FILE NAME:	PAGE 35 OF 52

WTR2955 POWER

REF 3100-3149



NOTE: C3119, C3109, C3107 ARE PRIORITY 1 AND SHOULD BE PLACED CLOSEST TO VDD PINS.

BASED ON 80-NL713-44 REV. A

MOTOROLA MOBILITY CONFIDENTIAL RESTRICTED-MDB TRACKED TRACKEDANDBLOCKED

APPROVALS		 MOTOROLA MOBILE DEVICES	
ENGINEER: MMI		BLOCK TITLE:	
LAST MODIFIED BY: MMI Fri Feb 19 14:26:31 2016		blk_wtr2955	
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		CAD FILE NAME:	
		PAGE 36 OF 52	

REF 3050-3099

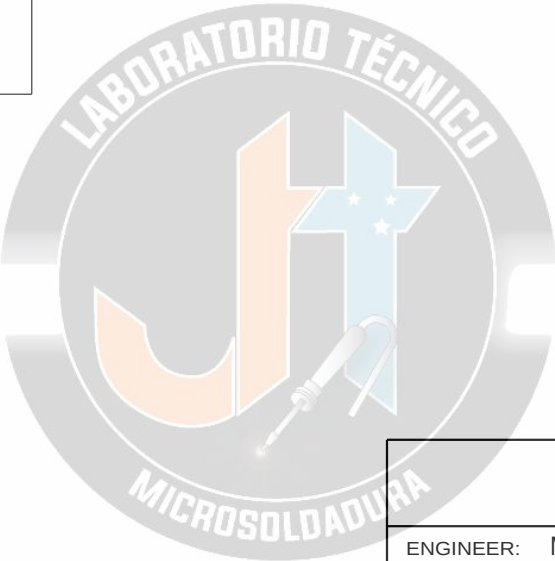
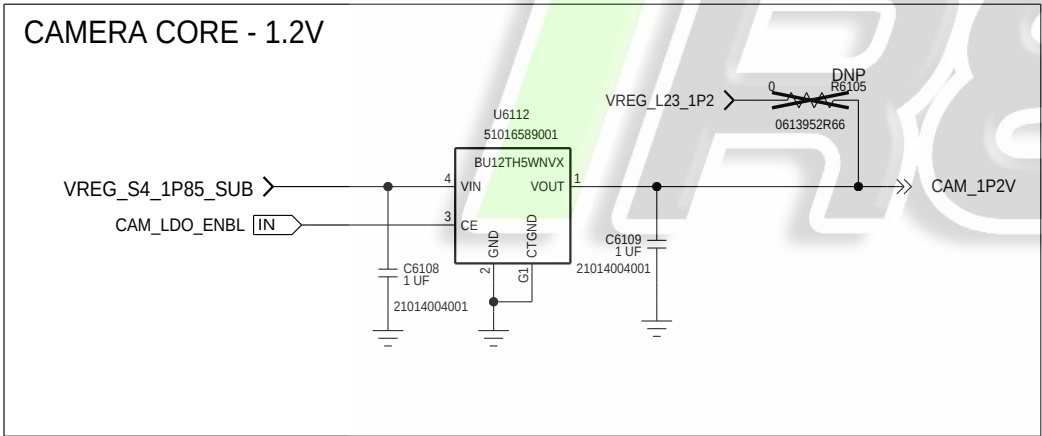
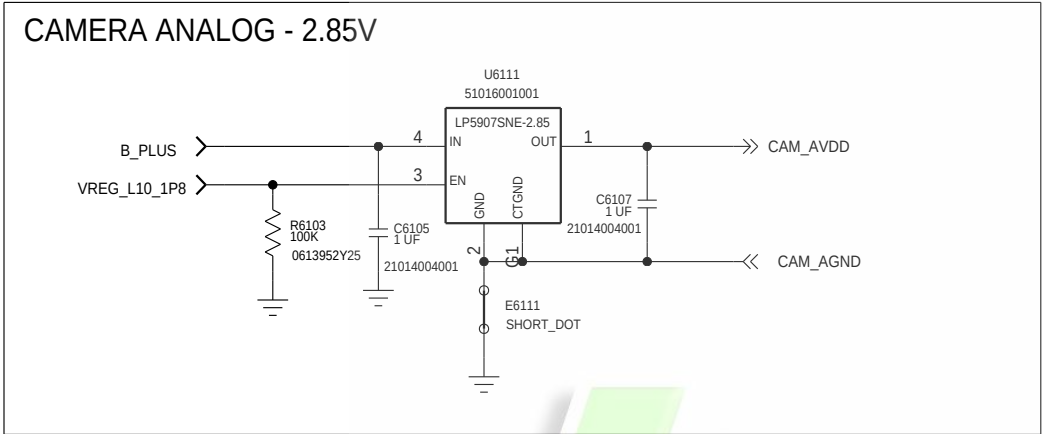


NOTE: DESIGN REQUIREMENTS FOR PRX_CA INTERCONNECTIONS AND DRX_CA INTERCONNECTIONS:

- TOTAL TRACE LENGTH BETWEEN WTR0 AND WTR1 PIN < 15 MM
- ISOLATION BETWEEN PRX_CA AND DRX_CA PINS > 50 DB
- KEEP TRACE IMPEDANCE 40 OHMS WITH GND ABOVE AND BELOW THESE TRACES

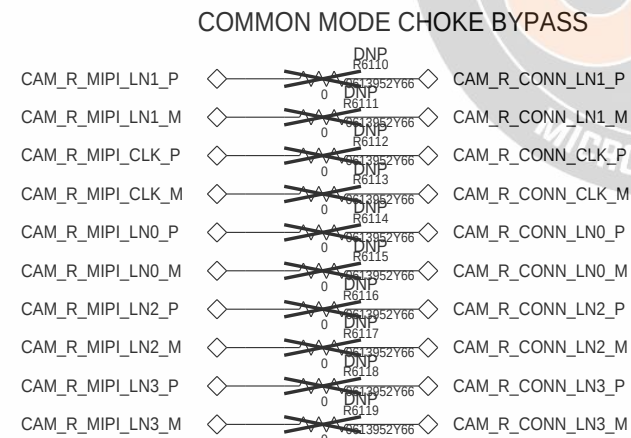
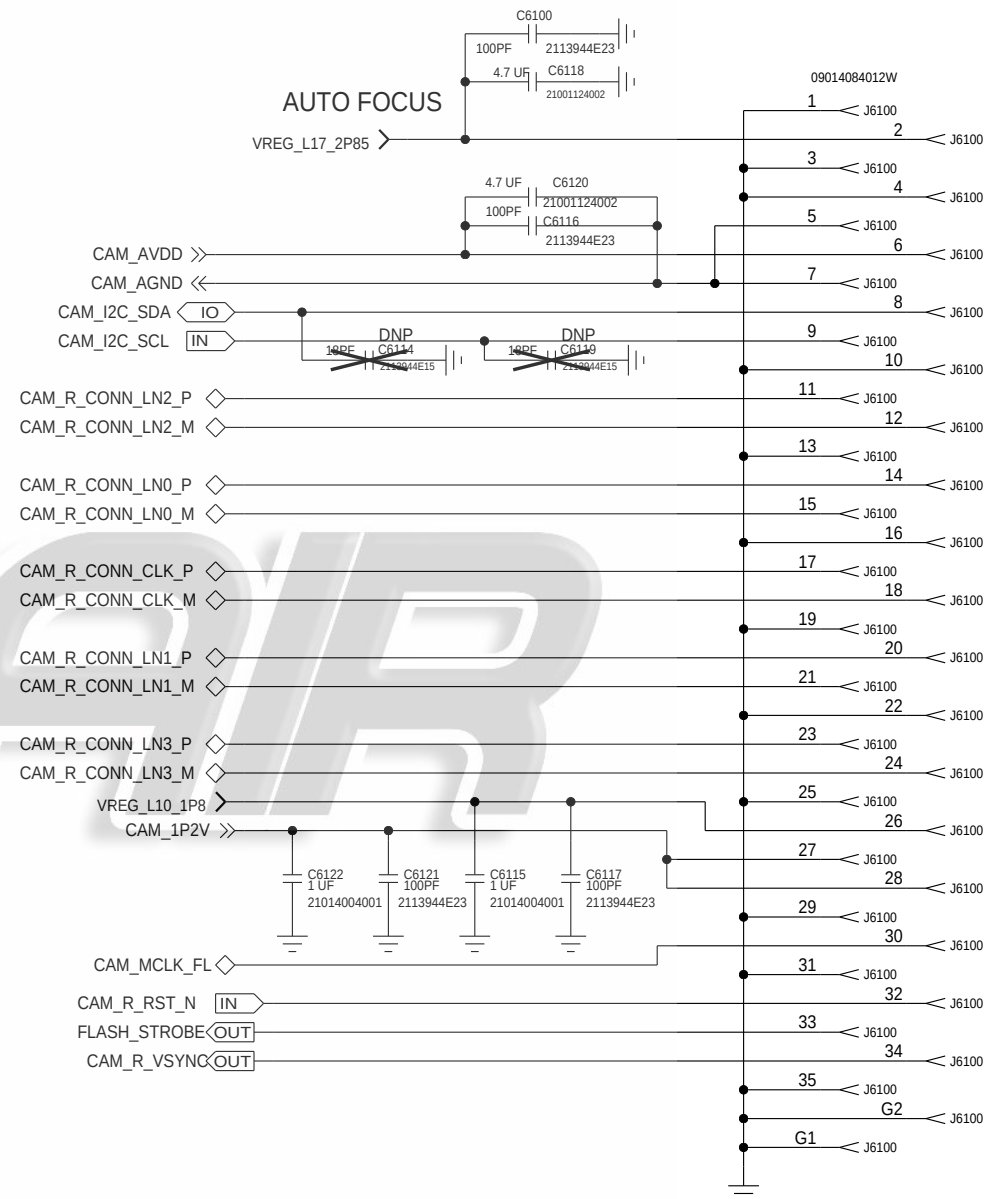
APPROVALS		 MOTOROLA MOBILE DEVICES	
ENGINEER: MMI	BLOCK TITLE: ca_wtr2955		
LAST MODIFIED BY: Fri Feb 19 14:26:32 2016			
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		CAD FILE NAME:	PAGE 37 OF 52

CAM: POWER



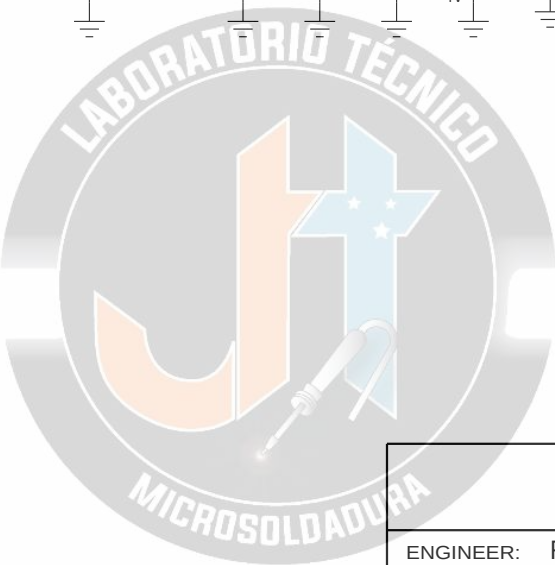
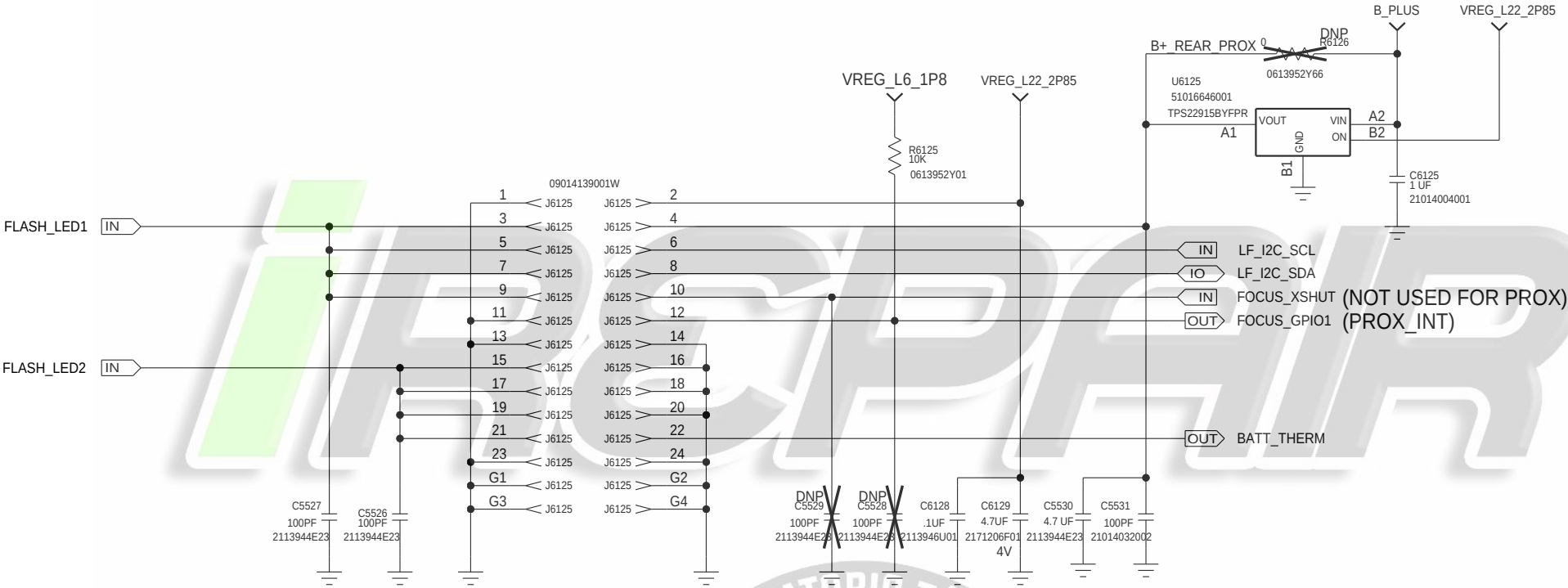
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		B 84016530001_a.cpm	REV 01
		CAD FILE NAME:	
		PAGE 39 OF 52	

REF:6100-6124



APPROVALS		 MOTOROLA MOBILE DEVICES	
ENGINEER: JOHN KRAMER	BLOCK TITLE: multimedia		
LAST MODIFIED BY: AZIZ NOORI Fri Feb 19 14:26:34 2016			
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		CAD FILE NAME:	PAGE 40 OF 52

LASER FOCUS & FLASH CONN



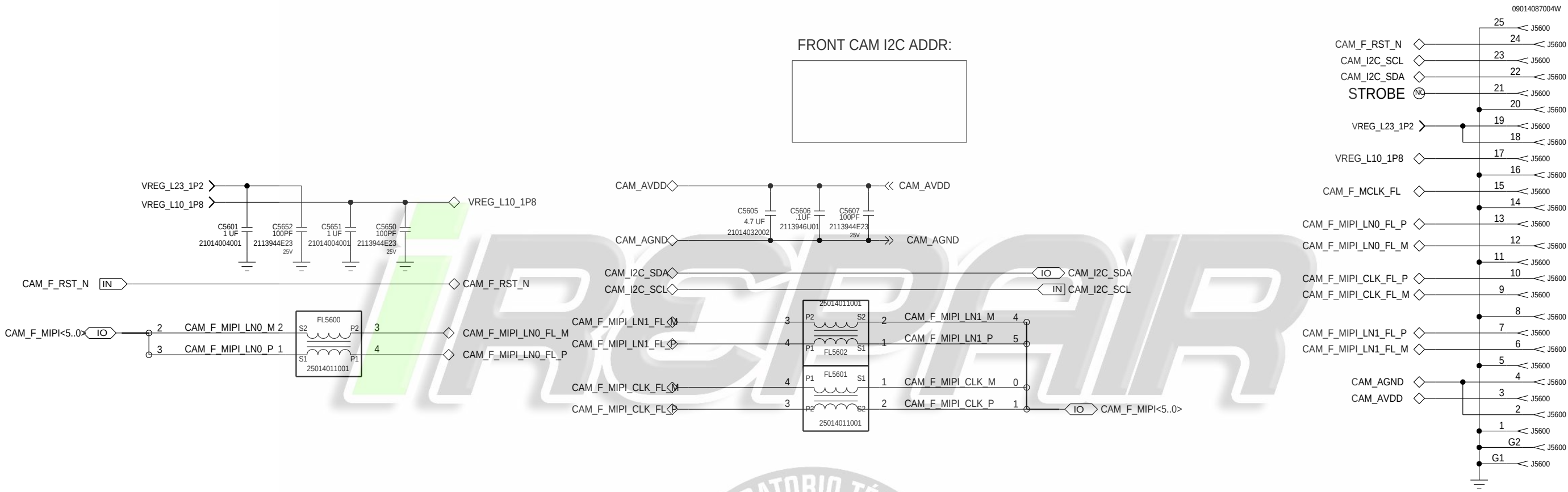
APPROVALS		MOTOROLA MOBILE DEVICES	
ENGINEER: RICH WOLNY		BLOCK TITLE: multimedia	
LAST MODIFIED BY: HUNG PHAM Fri Feb 19 14:26:36 2016			
MOTOROLA CONFIDENTIAL RESTRICTED THE DRAWING ON THIS PRINT AND INFORMATION THEREWITH ARE PROPRIETARY TO MOTOROLA AND SHALL NOT BE USED OR DISCUSSED IN WHOLE OR IN PART WITHOUT MOTOROLA'S CONSENT. © COPYRIGHT, MOTOROLA. THIS DOES NOT IMPLY PUBLICATION		B PROJECT NAME 84016530001_a.cpm	REV 01
		CAD FILE NAME: PAGE 41 OF 52	

CAM: FRONT

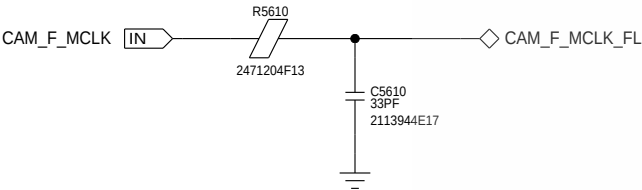
REF 5600-5649

FF CAMERA CONNECTOR

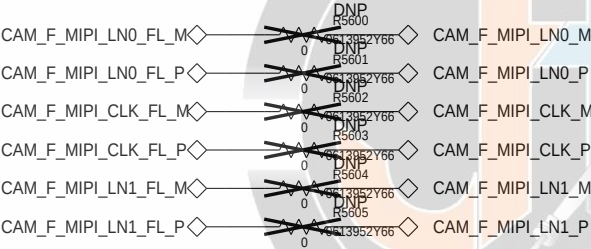
FRONT CAM I2C ADDR:



EMI/ESD FILTER

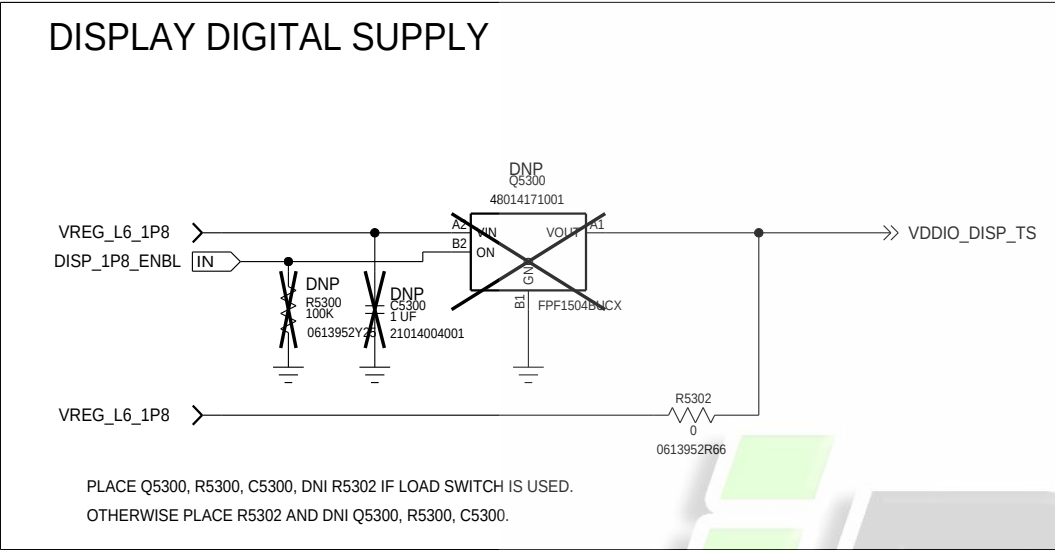


COMMON MODE CHOKE BYPASS

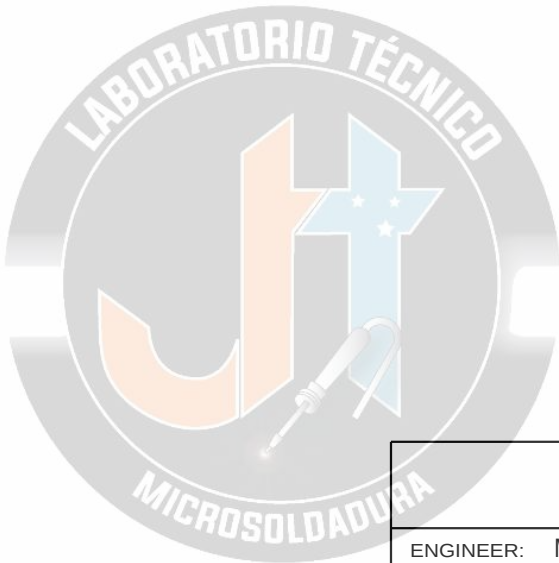


APPROVALS		MOTOROLA MOBILE DEVICES	
ENGINEER: JOHN KRAMER		BLOCK TITLE: multimedia	
LAST MODIFIED BY: NATE CONNELL Tue Feb 16 11:14:52 2016			
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		CAD FILE NAME: PAGE 42 OF 52	

DISP: POWER

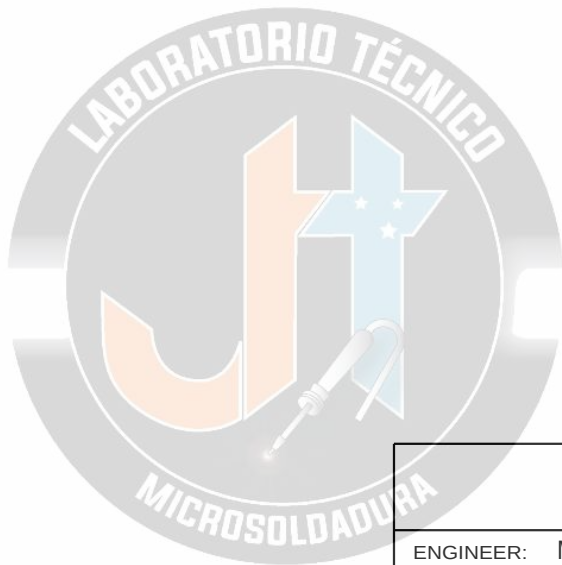
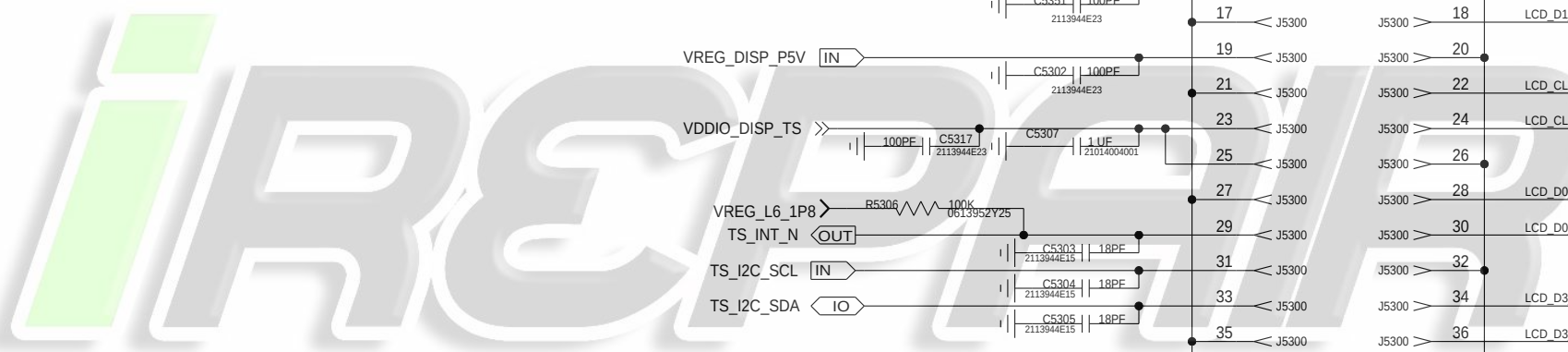


REPAIR

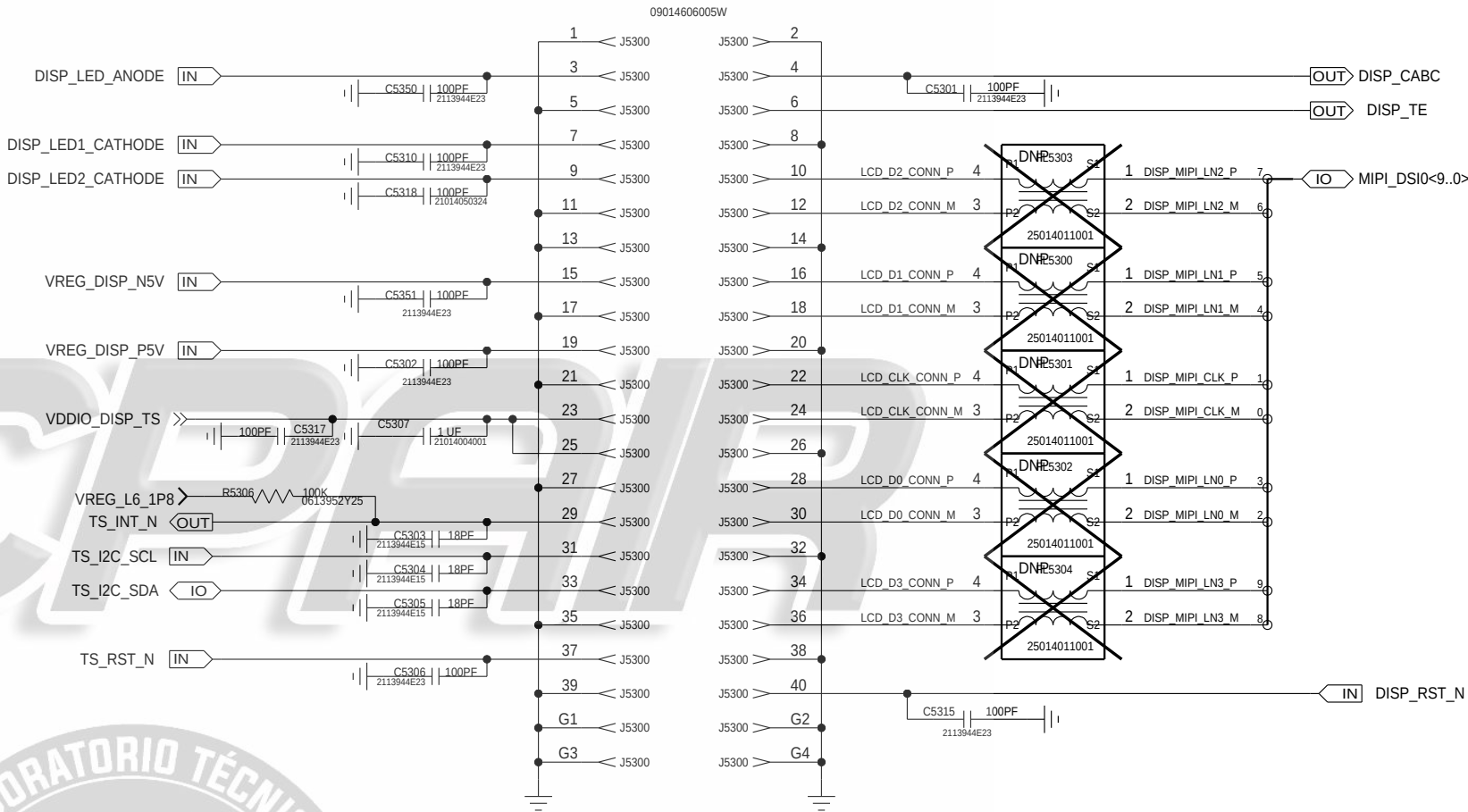
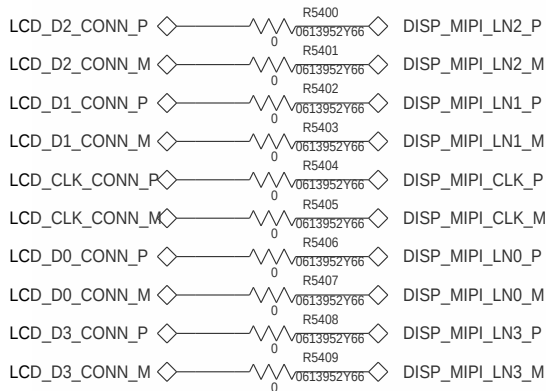


APPROVALS		 MOTOROLA MOBILE DEVICES		
ENGINEER: MMI		BLOCK TITLE: multimedia		
LAST MODIFIED BY: MMI Fri Feb 19 14:26:36 2016				
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		B 84016530001_a.cpm		01
		CAD FILE NAME:		PAGE 43 OF 52

DISP: CONNECTOR

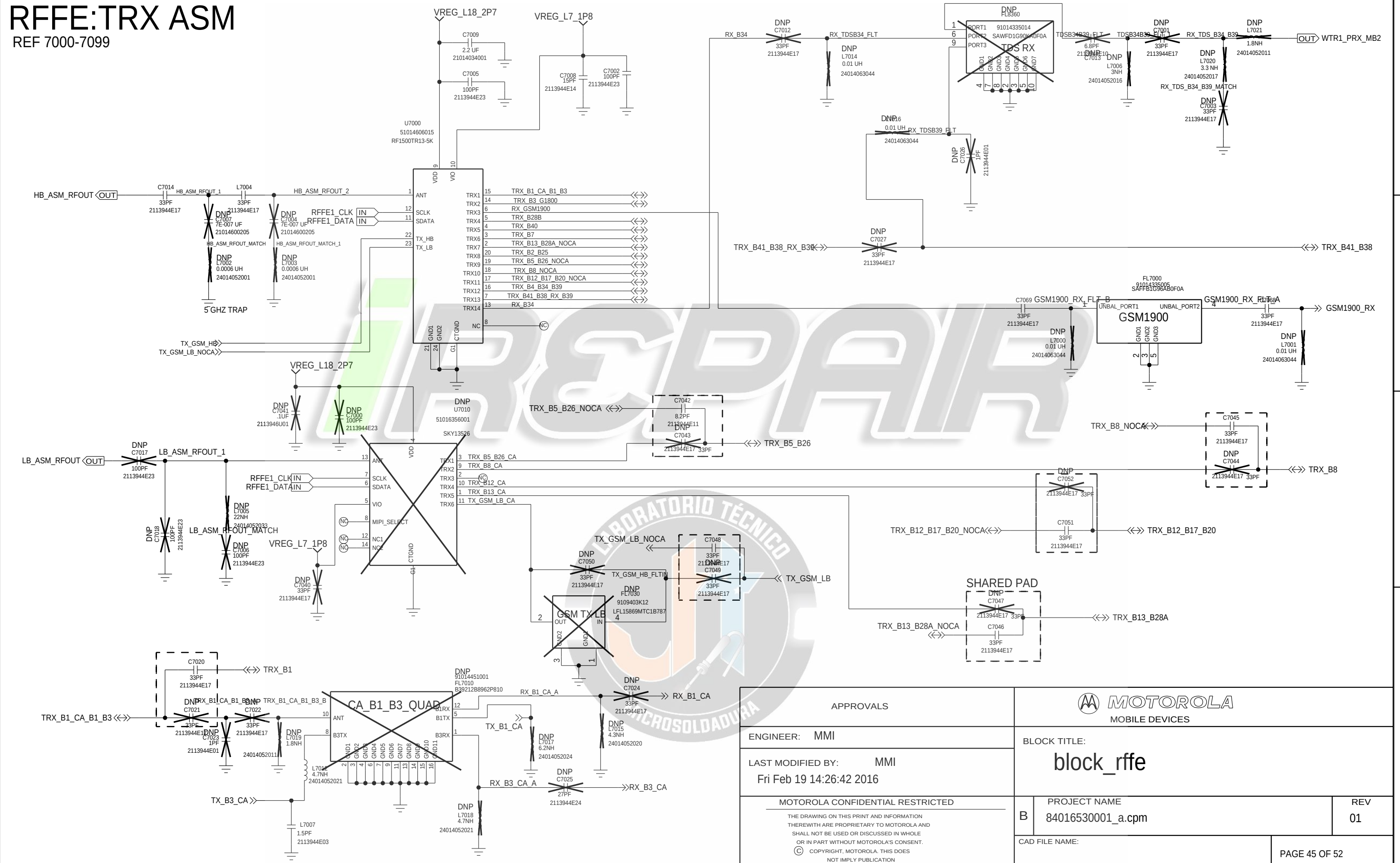


COMMON MODE CHOKE BYPASS



APPROVALS		MOTOROLA MOBILE DEVICES	
ENGINEER: MMI		BLOCK TITLE:	
LAST MODIFIED BY: MMI Fri Feb 19 14:26:35 2016		multimedia	
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		B 84016530001_a.cpm	REV 01
		CAD FILE NAME:	
		PAGE 44 OF 52	

REF 7000-7099



APPROVALS

ENGINEER: MMI

LAST MODIFIED BY: MMI

Fri Feb 19 14:26:42 2016

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BLOCK TITLE:

block_rffe

		PROJECT NAME
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B	84016530001_a.cpm
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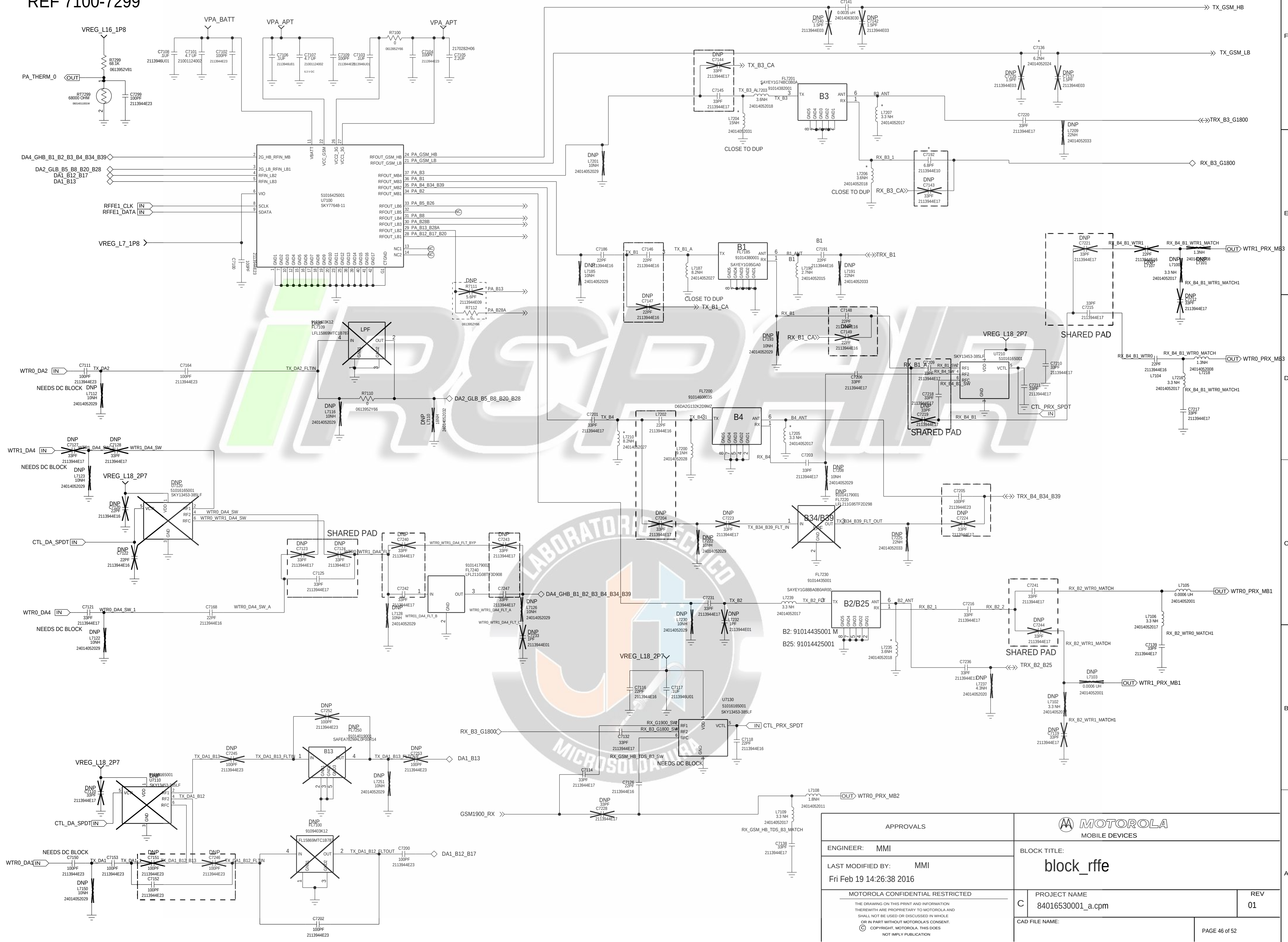
CAD FILE NAME:

EV

1

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REF 7100-7299

[illegible]

REF 7300-7399



C7323 2.2UF REMOVED WHEN C7343 COMMONLY USED AT LAYOUT
C7343 ALWAYS PLACED

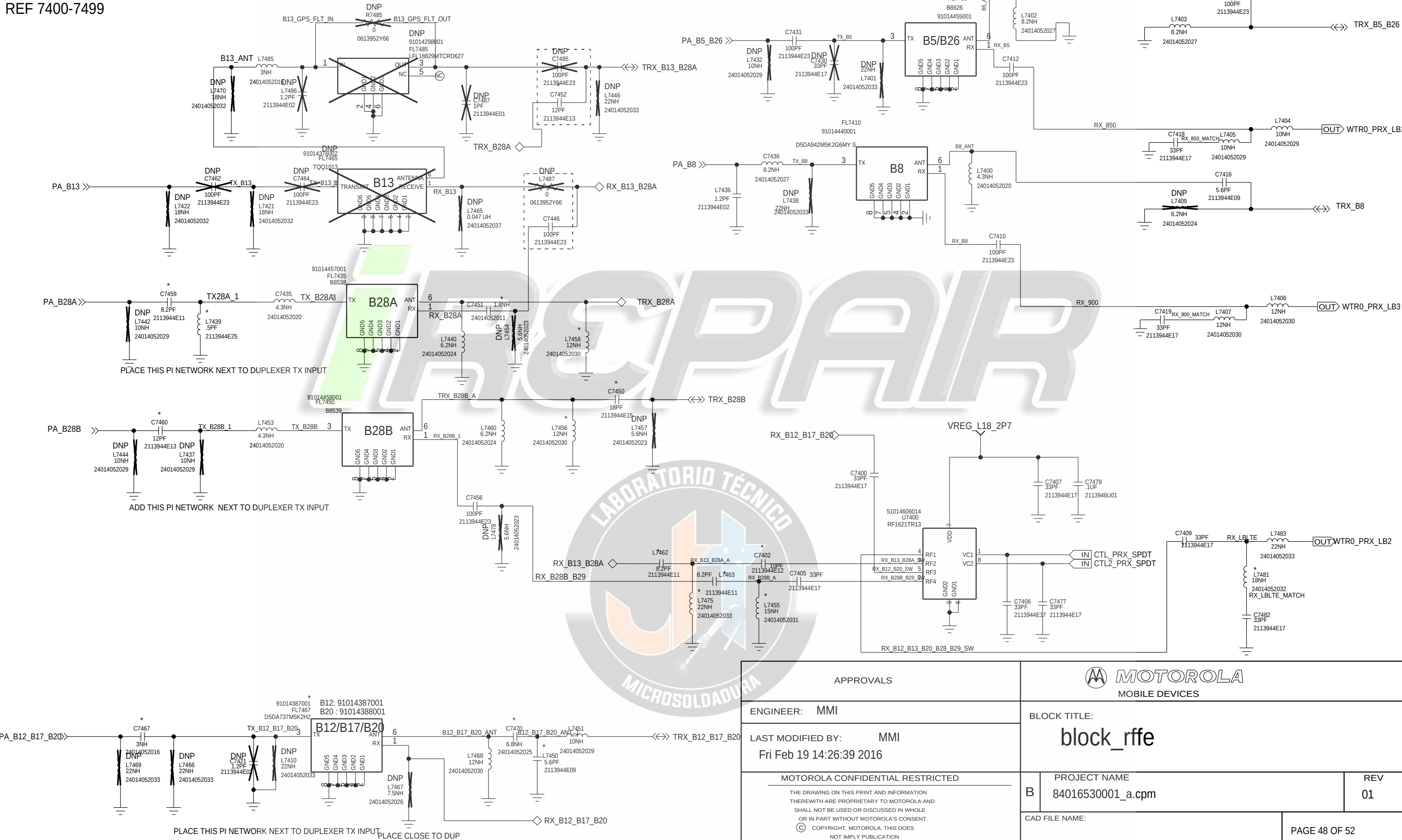
SHARED PAD

FOR MULTIPLE VENDOR CAPATIBILITY

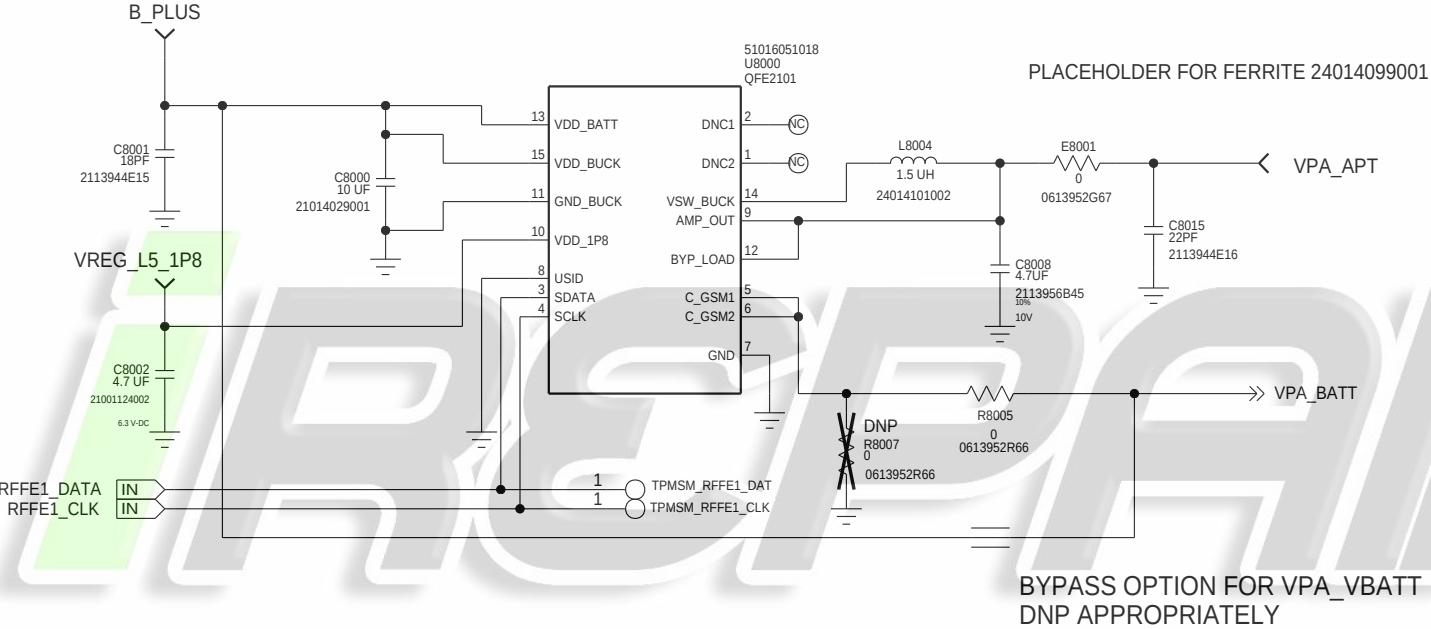
B7=SKY77781-11. RES REMOVED AT PIN8

APPROVALS		 MOTOROLA MOBILE DEVICES	
ENGINEER: MMI	BLOCK TITLE: block_rffe		
LAST MODIFIED BY: MMI Fri Feb 19 14:26:40 2016			
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		CAD FILE NAME:	PAGE 47 OF 52

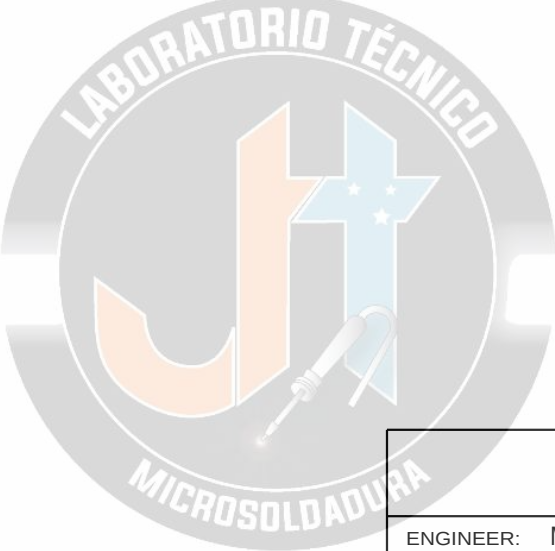
RFFE:LB TRX
REF 7400-7499



RFFE:QFE2101
REF 7400-7499

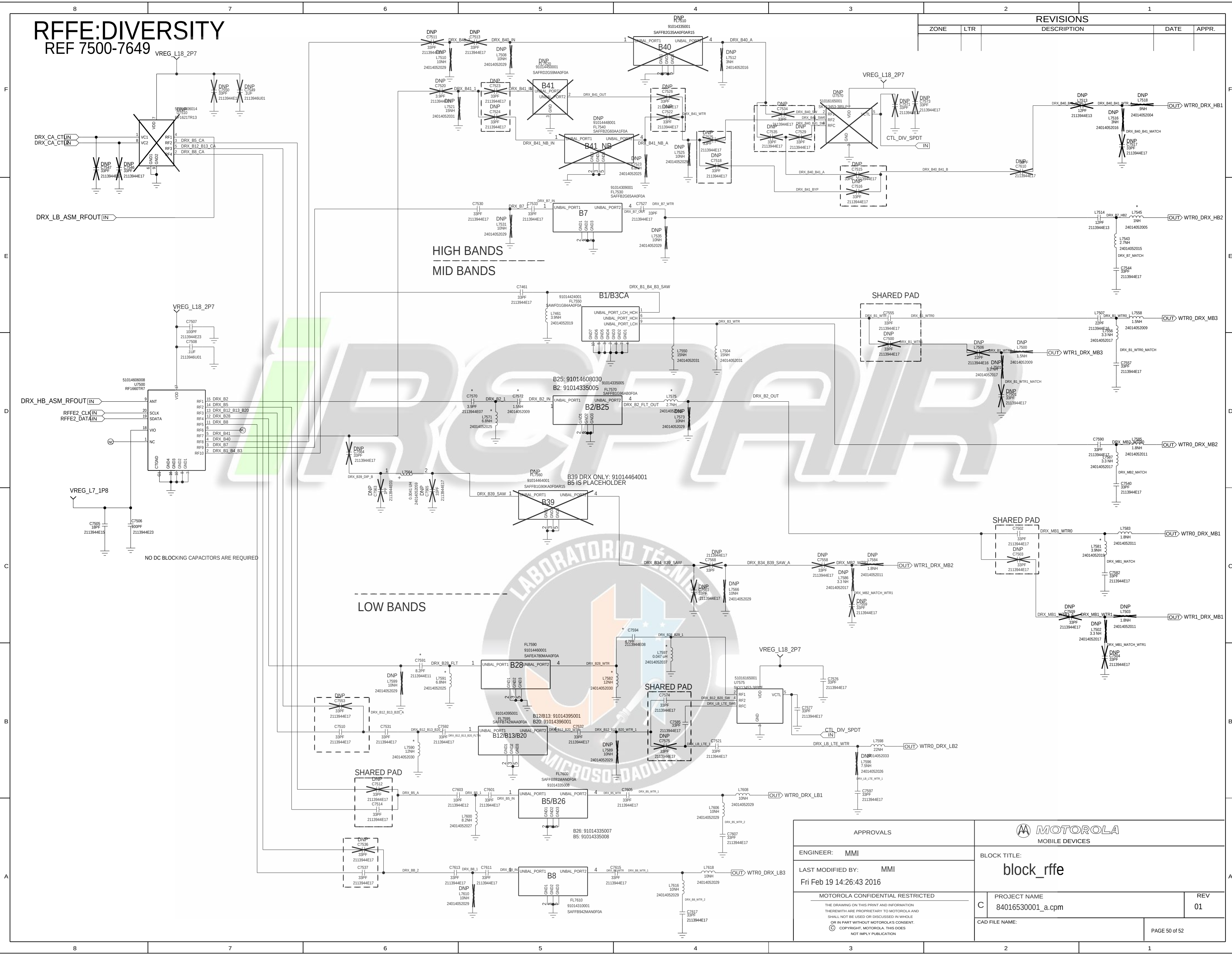


GND_BUCK: DO NOT VIA DIRECTLY TO GROUND PLANE. KEEP ISOLATED ON SURFACE LAYER;
ROUTE TO INPUT CAPACITOR GROUND PAD, AND THEN ONTO A VIA THAT CONNECTS TO THE GROUND
PLANE AT LEAST 1 MM BEYOND THE CAPACITOR.



APPROVALS		 MOTOROLA MOBILE DEVICES	
ENGINEER: MMI		BLOCK TITLE:	
LAST MODIFIED BY: MMI Fri Feb 19 14:26:37 2016		block_rffe	
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		CAD FILE NAME:	
		PAGE 49 OF 52	

RFFE:DIVERSITY
REF 7500-7649

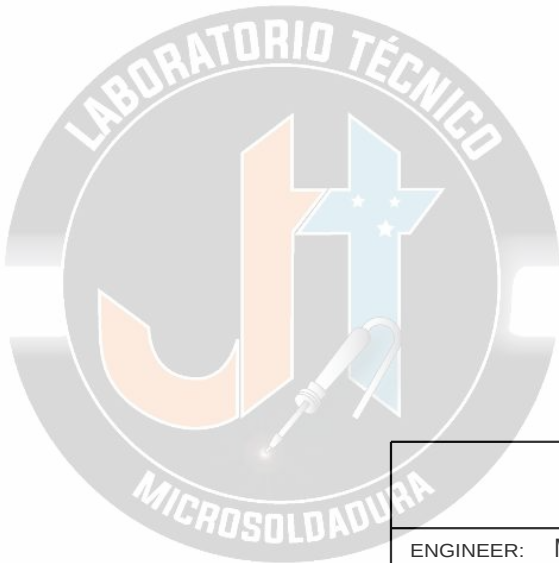
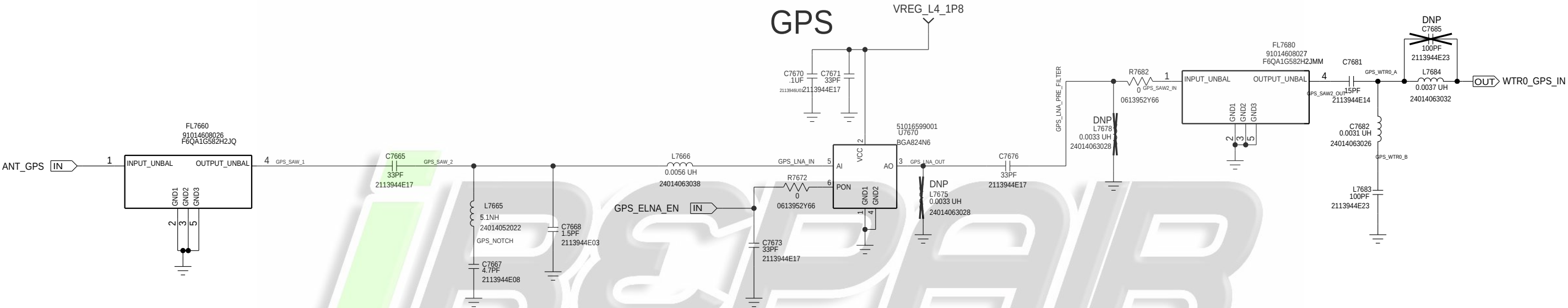


REVISIONS			
ZONE	LTR	DESCRIPTION	DATE

APPROVALS		MOTOROLA MOBILE DEVICES	
ENGINEER: MMI		BLOCK TITLE: block_rffe	
LAST MODIFIED BY: MMI		PROJECT NAME 84016530001_a.cpm	
Fri Feb 19 14:26:43 2016		REV 01	
MOTOROLA CONFIDENTIAL RESTRICTED		CAD FILE NAME:	
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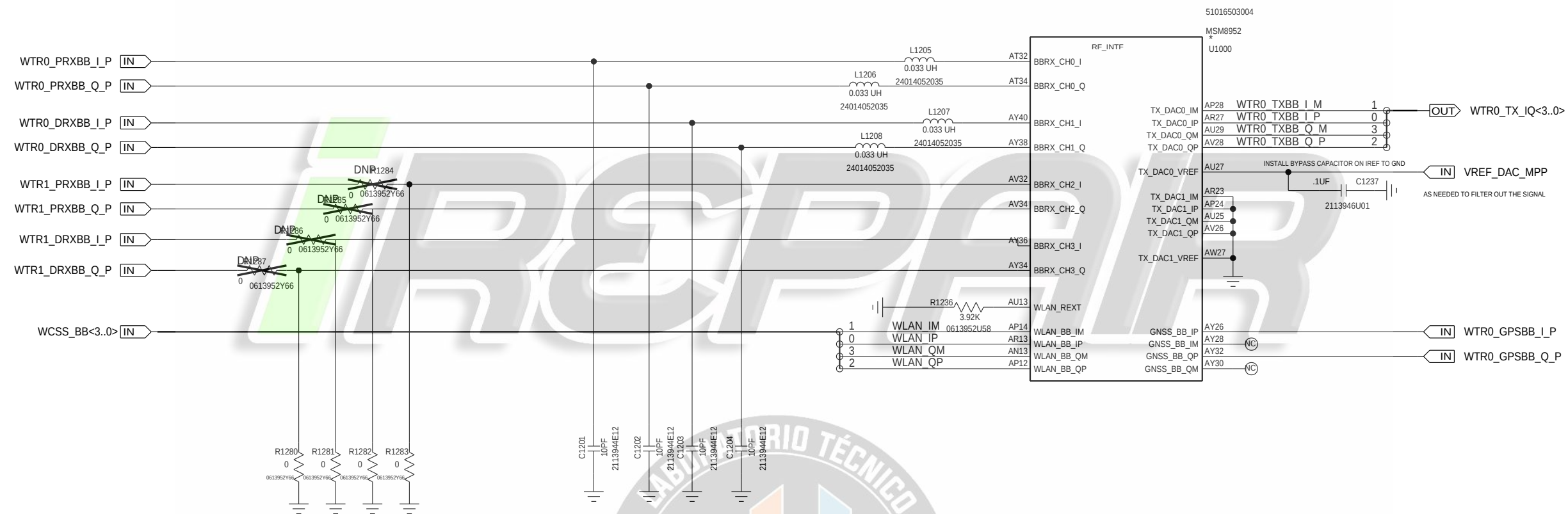
RFFE: GPS

REF 7650-7699



APPROVALS		MOTOROLA MOBILE DEVICES	
ENGINEER: MMI		BLOCK TITLE:	
LAST MODIFIED BY: MMI		block_rfe	
Web: 19-10-2013			
MOTOROLA CONFIDENTIAL RESTRICTED		PROJECT NAME	
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		CAD FILE NAME:	
		PAGE 51 OF 52	

MSM: RF INTERFACE
REF 1200-1299



APPROVALS		MOTOROLA MOBILE DEVICES	
ENGINEER:		BLOCK TITLE:	
LAST MODIFIED BY: FAN021 Fri Feb 19 14:26:45 2016		rf_intf	
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